



Wind Farm Feasibility Assessment for a Proposed Onshore Development in Ireland

Independent Technical Project:
Resource Assessment, Layout Optimisation,
Constraint Mapping and Energy Yield Analysis

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Executive Summary

This report presents a feasibility-stage assessment for a proposed onshore wind farm in Ireland, developed using a bespoke engineering workflow integrating geospatial screening, wind resource assessment, turbine layout optimisation, and wake-loss modelling. The objective was to replicate the core analytical stages of an early renewable energy development study using open-source tools while maintaining transparent and defensible engineering logic.

The project was undertaken in the context of limited access to commercial platforms such as WindPro. Rather than reducing scope, an alternative workflow was designed using public datasets, QGIS, Python automation, and the PyWake modelling framework. This enabled progression from national-scale screening to site selection, parcel assessment, candidate layout generation, energy yield comparison, and final concept refinement.

Five candidate Irish regions were first ranked using ERA5-derived wind resource indicators. Assessment metrics included mean wind speed at a height of 100 m, wind power density, Weibull parameters, winter wind performance, directional consistency, shear exponent, and a turbulence proxy. Galway ranked highest overall, demonstrating the strongest combined resource quality and was selected for detailed spatial screening.

A GIS constraints assessment was then completed within the selected region using exclusion layers representing key development limitations, including buildings, roads, water features, and other land-use constraints. Following screening, 214 individual land parcels remained. These parcels were classified by developable area and prioritised using a weighted scoring method favouring strong building separation and practical infrastructure access.

Parcel ID 7 emerged as the preferred primary development parcel, offering the best balance of scale, contiguity, and realistic layout potential. Candidate turbine layouts were tested within this parcel using a representative Vestas V136-4.2 MW turbine model. Results showed that a constrained 10-turbine arrangement provided the most efficient single-parcel solution, achieving 109.74 GWh net annual energy production with a wake loss of 5.22%.

To preserve the original benchmark project scale of 12 turbines, adjacent parcels were then evaluated for expansion potential. Five multi-parcel development options were assessed. The strongest-performing configuration, Candidate D, achieved 131.61 GWh net annual energy production at 50.4 MW installed capacity with a wake loss of 5.27%. Compared with the unconstrained 12-turbine benchmark, this refined layout retained almost all net energy yield, with only a 0.49% reduction while adapting to realistic land constraints.

Beyond the engineering results, the project demonstrates the ability to build a defensible analytical pipeline, adapt methodology under software constraints, interpret complex outputs, and translate technical findings into practical development decisions. These capabilities are directly relevant to modern renewable energy engineering roles requiring analytical judgement, geospatial problem-solving, optimisation, and clear communication.

The current project scope covers wind resource screening, GIS-based suitability analysis, layout optimisation, wake modelling, and comparative feasibility assessment. Planned future extensions to this project include grid integration screening, battery storage assessment, and full techno-economic evaluation to progress the concept from technical feasibility toward investment-grade project appraisal.

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Section 1. Introduction

1.1 Background

The global energy sector is undergoing a structural transition driven by decarbonisation targets, electrification of heat and transport, energy security concerns, and continued improvements in renewable generation technology. Among the available low-carbon technologies, onshore wind remains one of the most mature and commercially deployable options due to its proven performance, scalable project sizes, relatively short construction timelines, and declining lifetime cost of energy in suitable wind regimes.

Across Europe, wind energy has become a central component of national energy strategies. Ireland is particularly well positioned within this transition due to its strong Atlantic wind resource, established renewable policy framework, and increasing need for indigenous generation capacity. The combination of favourable meteorological conditions and long-term system decarbonisation requirements makes the Irish market an appropriate case study for applied wind farm development analysis.

Despite the strategic opportunity presented by wind generation, successful project delivery depends on far more than raw wind speed. Viable wind farm development requires the integration of resource assessment, land availability, environmental constraints, civil engineering practicality, turbine layout design, wake-loss management, grid access considerations, planning risk, and commercial performance. Decisions made during the early feasibility stage can materially influence whether a project progresses efficiently or hits barriers later in development.

1.2 Problem Context

Professional wind energy developers typically rely on specialist commercial software packages to support early-stage screening, energy yield estimation, micro-siting, and optimisation. Platforms such as WindPro, OpenWind, and related proprietary tools provide mature workflows for these tasks. However, access to such software can be limited by licensing cost, institutional availability, or project context.

This creates an important practical challenge. How can a technically credible feasibility assessment still be completed when ideal commercial tools are unavailable?

That challenge formed the basis of this project. Rather than treating software access as a barrier, the study was approached as an engineering problem requiring method selection, transparent assumptions, and reproducible analysis. Publicly available datasets, open-source geospatial tools, and Python-based modelling methods were combined to create a defensible development workflow capable of supporting real decision-making at desk-study level.

1.3 Purpose of the Study

The purpose of this report is to document the end-to-end development of a feasibility-stage onshore wind farm assessment for Ireland, progressing from national-scale site screening to refined turbine layout comparison.

The study was designed to answer five core questions:

1. Which candidate region offered the strongest balance of wind resource quality and development potential?
2. After applying practical land-use constraints, which parcels of land remained suitable for development?
3. What turbine capacity could be realistically accommodated within the preferred parcel?
4. Could neighbouring land parcels be used strategically to recover benchmark project scale?
5. How closely could a constrained real-world concept match the performance of an unconstrained benchmark layout?

These questions reflect the type of staged decision-making process commonly encountered during real project origination and pre-development screening.

1.4 Project Approach

To address the above objectives, a multi-stage analytical workflow was developed. The overall methodology adopted in this study is summarised in Figure 1, outlining the progression stages shown below:

- regional wind resource screening using ERA5 reanalysis data;
- weighted ranking of five candidate Irish regions;
- GIS-based exclusion mapping and land suitability filtering;
- parcel-level viability classification and prioritisation;
- turbine technology selection for feasibility modelling;
- candidate layout generation under spacing and boundary constraints;
- wake-loss and annual energy production assessment using PyWake;
- comparison of refined constrained layouts against an initial benchmark concept.

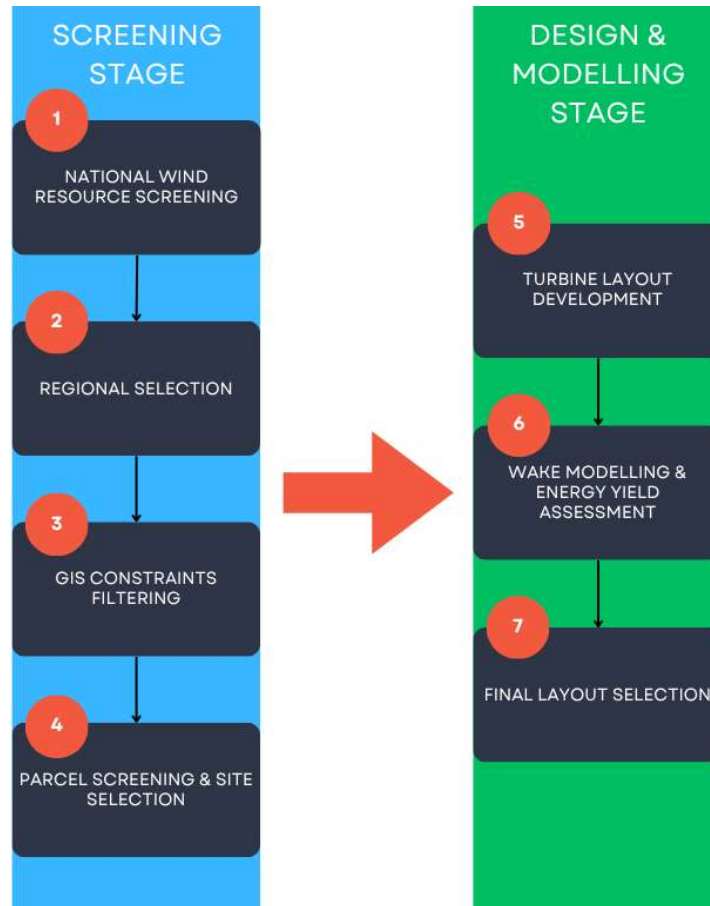


Figure 1 - Current Project Workflow

The project therefore combines geospatial analysis, atmospheric data interpretation, engineering judgement, numerical modelling, and decision-focused reporting.

1.5 Professional Relevance

In addition to the technical objectives, the project was designed to demonstrate several transferable engineering capabilities relevant to modern renewable energy roles.

These include:

- designing analytical workflows under practical constraints;
- adapting methodology where preferred tools are unavailable;
- integrating multiple data sources into a coherent decision-making process;
- translating model outputs into development recommendations;
- communicating technical findings clearly to mixed audiences;
- maintaining traceability between assumptions, methods, and conclusions.

For that reason, this report is intended not only as a technical case study, but also as a demonstration of applied engineering problem solving in a real-world project context.

1.6 Report Structure

The remainder of this report follows the project development sequence from high-level screening to detailed feasibility assessment. Subsequent sections present the data sources used, regional ranking methodology, GIS constraint filtering, parcel selection logic, turbine and layout design methods, wake-modelling results, engineering interpretation, project limitations, and recommended next steps. The project stages and associated outputs are summarised in Table 1.

Table 1 - Summary of Project Stages and Outputs

Stage	Project Phase	Primary Tools / Methods	Key Inputs	Principal Outputs
1	National Wind Resource Screening	ERA5, Python, statistical analysis	Candidate Irish regions, historical wind variables, coordinates	Comparative wind resource dataset, ranked candidate regions
2	Regional Selection	Weighted scoring framework, Python	Normalised screening metrics, engineering weightings	Preferred study region selected
3	GIS Constraints Filtering	QGIS, geospatial overlay analysis	Study boundary, buildings, roads, hydrology, environmental layers, terrain constraints	Screened developable land bank, residual unconstrained polygons
4	Parcel Screening & Site Selection	QGIS, Python, spatial metrics	Parcel geometries, parcel area, distance to roads, distance to buildings	Parcel rankings, viability classes, preferred parcel
5	Turbine Layout Development	Python, engineering spacing logic, geometric iteration	Parcel boundary, turbine dimensions, spacing rules, benchmark layout assumptions	Candidate turbine coordinate layouts (5T-12T), preferred internal 10T layout
6	Wake Modelling & Energy Yield Assessment	PyWake, Python	Turbine layouts, wind climate inputs, Weibull parameters, directional frequencies, turbine curves	Gross AEP, net AEP, wake loss metrics, comparative layout performance
7	Final Layout Selection	Comparative engineering assessment	Ranked modelling outputs, spacing metrics, benchmark comparison	Selected final layout:
8	Future Project Expansion	Planned engineering studies	Final technical layout, project assumptions, network context	Grid integration review, storage options, techno-economic assessment

Section 2. Data Sources and Study Inputs

2.1 Overview

The quality of any feasibility-stage wind farm assessment is strongly influenced by the suitability, transparency, and consistency of the input data used. This project combined meteorological datasets, geospatial screening layers, engineering assumptions, and internally generated optimisation outputs to create a traceable evidence base for decision-making.

Because the study was undertaken without a fully integrated commercial development platform, particular emphasis was placed on documenting assumptions and maintaining clear links between raw inputs, intermediate processing steps, and final engineering conclusions.

This section summarises the principal datasets, model assumptions, and engineering inputs used throughout the project.

2.2 Wind Resource Data

Regional wind screening and wake-modelling inputs were derived from ERA5 reanalysis data. ERA5 is a globally recognised atmospheric dataset produced through numerical weather prediction data assimilation and is widely used for preliminary renewable energy assessments where measured site data is not yet available.

Within this study, ERA5-derived metrics were used both to compare candidate Irish regions and to parameterise feasibility-stage wake simulations.

Primary wind variables used in the project included:

- mean wind speed at 100 m hub-height proxy;
- mean wind power density;
- Weibull scale parameter (c);
- Weibull shape parameter (k);
- winter mean wind speed;
- directional frequency distribution;
- shear exponent proxy;
- turbulence proxy indicator.

While reanalysis data cannot replace bankable on-site measurement campaigns, it provides a robust basis for comparative screening and early concept development when interpreted appropriately. The geographic locations of the candidate regions used for ERA5 data extraction are shown in Figure 2 and detailed in Table 2.

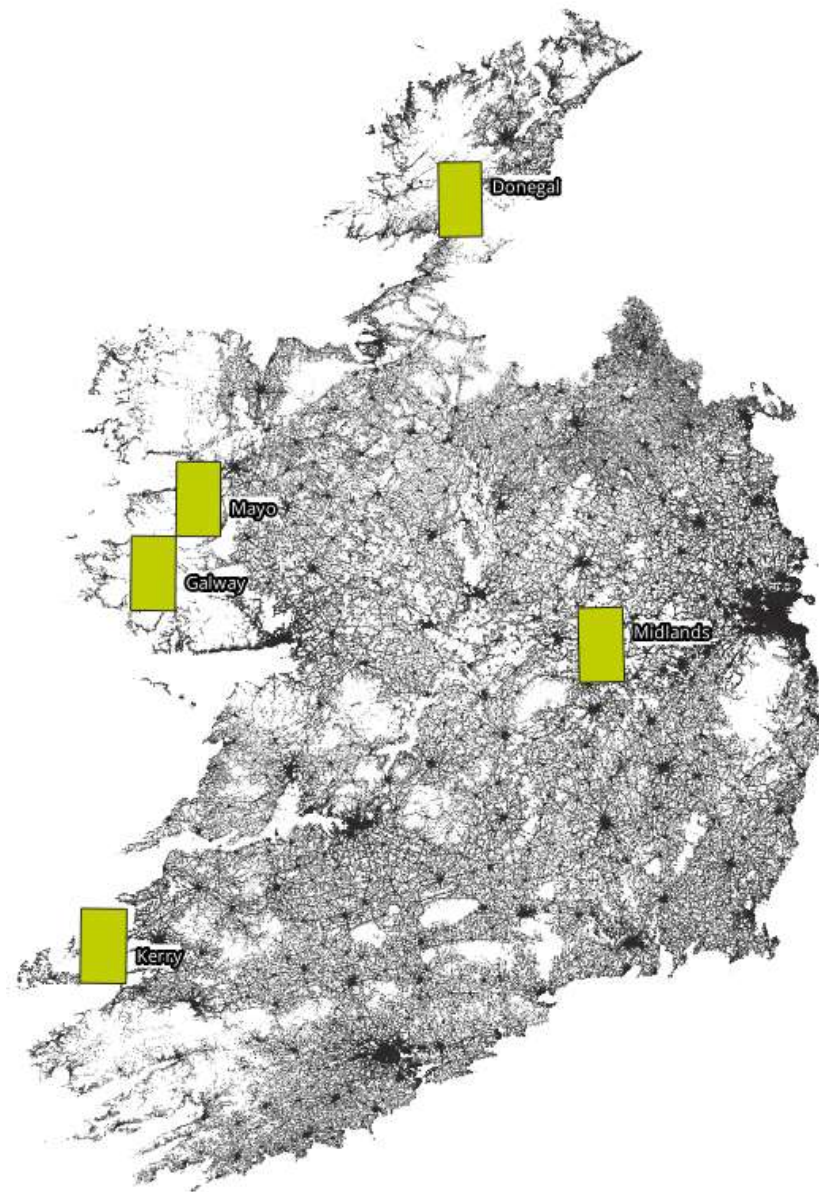


Figure 2 - ERA5 grid cells ($0.25^{\circ} \times 0.25^{\circ}$ resolution) representing the five candidate regions used during national wind resource screening.

Table 2 - Candidate Regional Wind Resource data computed from ERA5 wind time series to characterise the statistical wind regime at each candidate location.

Region	Latitude ($^{\circ}$ N)	Longitude ($^{\circ}$ W)	Mean Wind Speed at 100 m (m/s)	Weibull c (m/s)	Weibull k (-)
Galway	53.5	-9.75	7.91	8.94	2.21
Donegal	54.75	-8.0	7.2	8.12	2.29
Mayo	53.75	-9.5	7.23	8.17	2.23
Midlands	53.25	-7.25	6.68	7.53	2.43
Kerry	52.25	-10.0	7.21	8.14	2.16

2.3 Geospatial Constraint Data

Following regional selection, a GIS-based suitability assessment was undertaken to identify land remaining after exclusion of key development constraints. Multiple geospatial layers were combined in QGIS to progressively remove unsuitable land and isolate feasible parcels for further engineering review.

Constraint categories considered in the desk study included:

- building proximity buffers;
- road network proximity and access considerations;
- water features and drainage constraints;
- environmental or ecological exclusions where applicable;
- slope or terrain-related restrictions included during screening;
- boundary fragmentation effects following overlay operations.

The output of this stage was a constrained land suitability layer subsequently dissolved into individual developable parcels.

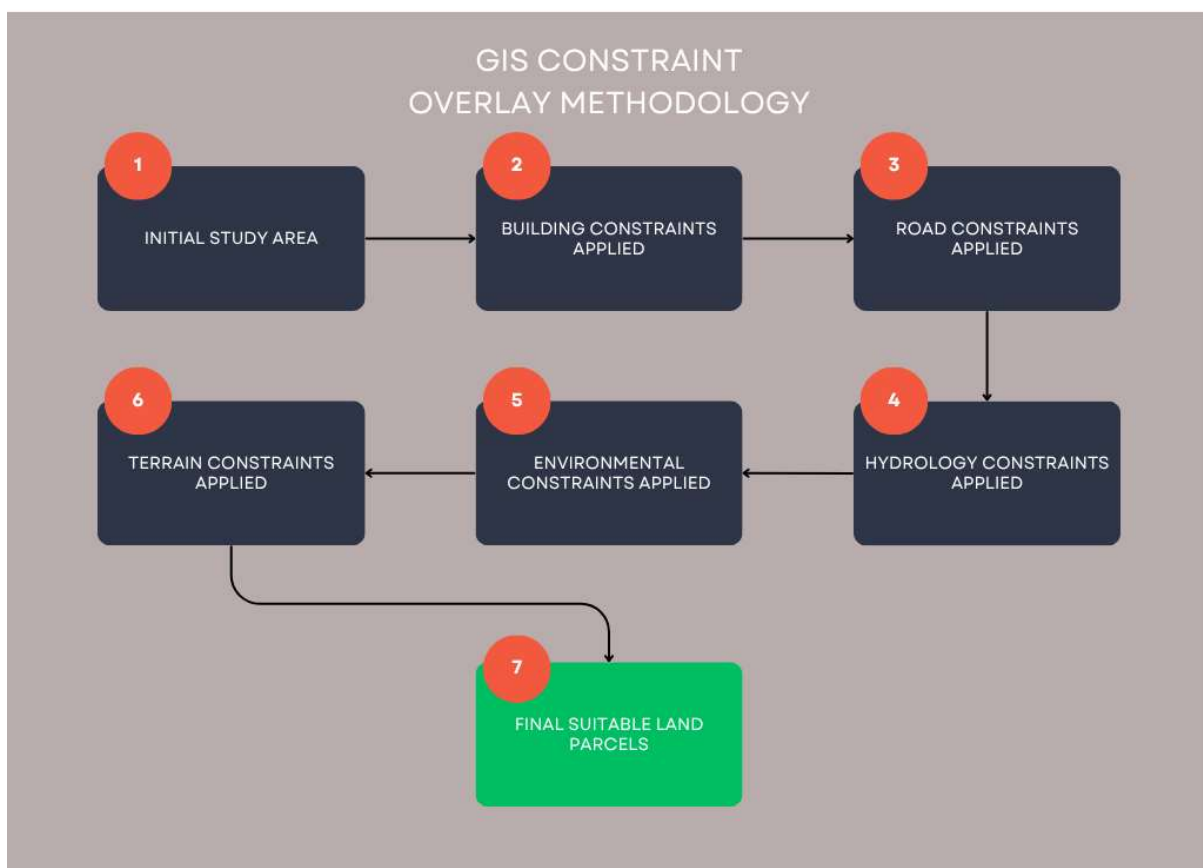


Figure 3 - GIS constraint overlay methodology: Illustrative process showing how the initial study area was progressively reduced through sequential application of building, road, hydrology, environmental, and terrain constraints to identify the final suitable land parcels for parcel-level screening.

2.4 Parcel Screening Dataset

The final suitability layer was converted into a parcel-level screening dataset. For each remaining parcel, geometric and locational metrics were calculated to support prioritisation and shortlist development candidates.

Key parcel attributes included:

- feature identifier (FID);
- distance to nearest building;
- distance to nearest road;
- area in m², ha, and km²;
- viability class based on area threshold;
- priority score derived from access and setback logic.

A total of 214 parcels remained after screening and formed the candidate land bank for detailed review.

2.5 Turbine Technical Inputs

Wake modelling and layout comparisons were undertaken using a representative Vestas V136-4.2 MW turbine specification selected as a realistic modern onshore reference machine for feasibility analysis.

Key model parameters were:

- rated power: 4.2 MW;
- rotor diameter: 136 m;
- hub height: 112 m;
- cut-in wind speed: 3 m/s;
- rated wind speed: 12 m/s;
- cut-out wind speed: 25 m/s.

Representative proxy power and thrust curves were used for modelling purposes. These curves were suitable for comparative feasibility assessment but should not be interpreted as OEM-certified performance guarantees.

The selected turbine specification provided a credible basis for comparing turbine counts, spacing behaviour, installed capacity, and annual energy output across candidate layouts.

2.6 Layout and Modelling Inputs

In addition to site and turbine data, the project required a set of derived modelling inputs generated during the analysis to support layout testing and wake simulation. These included:

- parcel boundaries used as optimisation envelopes;
- minimum spacing assumptions expressed in rotor diameters;
- candidate turbine coordinate sets;
- benchmark unconstrained reference layout;
- candidate multi-parcel expansion options A–E;
- wind rose sector probabilities for wake modelling;
- output directories and metadata logs for reproducibility.

These derived inputs formed the link between GIS-based screening outputs and subsequent layout optimisation and comparative performance modelling

2.7 Data Quality, Limitations, and Suitability

All inputs used in this study were selected to be appropriate for feasibility-stage assessment rather than final investment decision support. That distinction is important.

The dataset suite was sufficient to compare candidate opportunities, rank options consistently, and identify a preferred development concept. However, later project phases would require higher-resolution and project-specific inputs, including:

- on-site wind measurements;
- detailed topographic and roughness modelling;
- ground investigation data;
- ecological and planning surveys;
- electrical network capacity studies;
- supplier quotations and commercial cost data.

Accordingly, the results presented in this report should be interpreted as decision-support outputs for early-stage development rather than final design certification values.

Section 3. Wind Resource Assessment Methodology

3.1 Purpose of Regional Screening

Before detailed land suitability analysis can be justified, a project must first establish that the broader region offers an attractive wind resource. It is inefficient to commit significant effort to parcel screening, layout design, and energy modelling in locations where the underlying meteorological conditions are materially weaker than alternative opportunities.

For that reason, the first analytical stage of this project focused on comparative regional screening across multiple Irish candidate areas. The objective was not to determine bankable site yields at this stage, but to identify which region provided the strongest overall combination of wind energy potential and development relevance for further study.

3.2 Candidate Regions Assessed

Five candidate regions were selected to provide geographic spread and represent different Irish wind regimes:

- Galway
- Donegal
- Mayo
- Midlands
- Kerry

The selected set included western Atlantic-exposed counties with strong expected wind climates, together with an inland comparator region where wind speeds may be lower but land-use and development characteristics can differ.

Candidate locations were represented by coordinate points used for consistent ERA5 metric extraction. The coordinates and rationale used are summarised in Table 3 and depicted in Figure 4.

Table 3 - Candidate region coordinates used for screening. Geographic coordinates of the five Irish candidate regions selected for comparative wind resource assessment, together with the strategic rationale for inclusion in the regional screening stage.

Region	Latitude	Longitude	Screening Rationale
Galway	53.5	-9.75	Strong Atlantic wind exposure and selected final case
Donegal	54.75	-8	North-west coastal high-wind candidate
Mayo	53.75	-9.5	Western comparator with strong resource potential
Midlands	53.25	-7.25	Inland comparator with different land-use context
Kerry	52.25	-10	South-west candidate providing geographic diversity

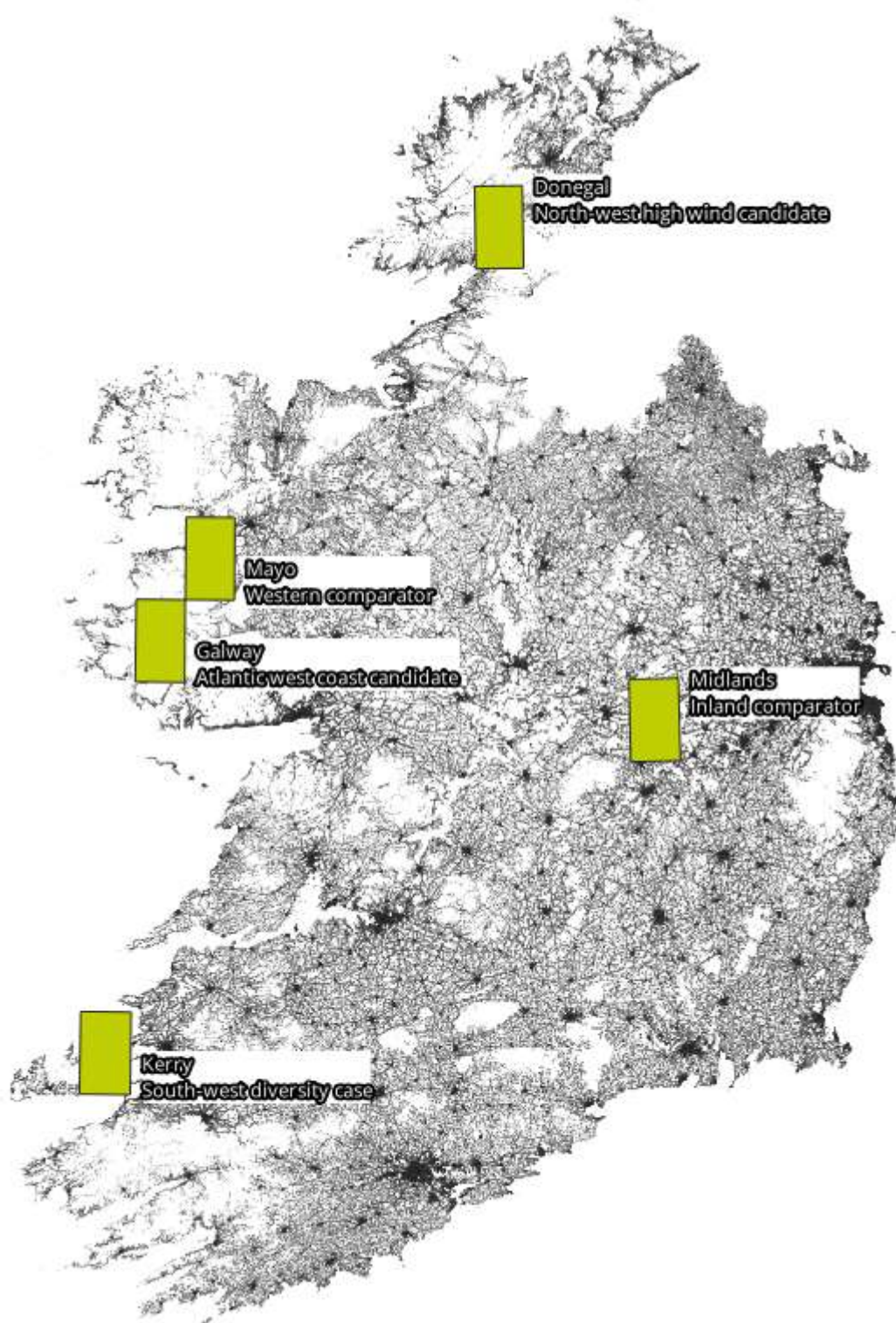


Figure 4 - Candidate regions selected for comparative wind resource assessment, showing geographic distribution and strategic rationale for inclusion.

3.3 Use of ERA5 Reanalysis Data

ERA5 reanalysis data was selected as the meteorological basis for screening due to its broad spatial coverage, internal consistency, and widespread use in early-stage renewable energy studies.

Reanalysis datasets combine numerical weather prediction models with assimilated observations to reconstruct historical atmospheric conditions across regular spatial grids. Although they do not replace site-specific measurement campaigns, they are particularly useful for comparative screening because all candidate regions are assessed under a common methodology.

This consistency is especially valuable during the option-selection stage, where relative ranking is often more important than exact absolute yield.

3.4 Screening Metrics Considered

Multiple wind metrics were evaluated rather than relying on mean wind speed alone. This reflects real development practice, where project attractiveness depends on both energy magnitude and operational characteristics.

Table 4 - Definition and engineering relevance of screening metrics. Summary of the principal ERA5-derived metrics used in regional wind resource screening, including their technical meaning, project relevance, and whether higher or lower values are generally preferred during early-stage site comparison.

Screening Metric	Definition	Why It Matters in Project Development	Generally Favourable Direction
Mean Wind Speed at 100 m	Average wind speed at 100 m hub-height proxy level.	Primary indicator of generation potential and expected annual energy yield. Higher average wind speeds generally increase turbine output.	Higher
Mean Wind Power Density	Average kinetic power available in the wind per unit swept area (W/m^2).	Captures the cubic relationship between wind speed and energy content, providing a stronger indicator of resource quality than wind speed alone.	Higher
Weibull Scale Parameter (c)	Statistical parameter representing characteristic wind speed of the site.	Higher values typically indicate a stronger overall wind climate and improved production potential.	Higher
Weibull Shape Parameter (k)	Statistical parameter describing the spread of the wind speed distribution.	Higher values generally indicate a more stable and predictable wind regime, supporting forecasting and operational consistency.	Higher
Winter Mean Wind Speed	Average wind speed during winter months.	Strong winter winds can be commercially valuable where electricity demand and market prices are seasonally elevated.	Higher

Directional Consistency	Degree to which wind frequencies are concentrated in dominant sectors.	Consistent prevailing directions support better turbine row alignment and reduced wake interaction through informed layout design.	Higher
Shear Exponent Proxy	Indicator of how wind speed changes with height above ground level.	Influences hub-height extrapolation, turbine suitability, and uncertainty in vertical wind profile assumptions. Moderate values are typically preferred.	Context dependent
Turbulence Proxy	Relative indicator of wind speed variability and flow disturbance.	Higher turbulence can increase structural fatigue loading, reduce power stability, and affect turbine lifetime performance.	Lower
Standard Deviation of Wind Speed	Statistical spread of wind speed values around the mean.	Helps indicate variability of the wind regime and complements turbulence interpretation during screening.	Lower / Moderate
Extreme Wind Speed Count	Count of unusually high wind speed events above a defined threshold.	Useful for understanding exposure to shutdown conditions, loading risk, and survivability considerations.	Lower

3.5 Weighted Scoring Framework

Because no single metric fully determines site quality, a weighted multi-criteria scoring framework was applied. Individual ERA5-derived metrics were first converted into comparative scores across the five candidate regions so that variables with different units and value ranges could be assessed on a consistent basis.

For metrics where higher values are generally favourable (such as wind resource strength, winter wind speed, Weibull shape factor, and directional consistency), higher raw values received higher comparative scores. For metrics where lower values are preferable (such as the turbulence proxy), the scoring scale was reversed. The shear exponent was assessed separately using an engineering banded score to reflect the preference for moderate shear conditions rather than simply maximising or minimising the raw value.

The resulting component scores were multiplied by their predefined weighting factors and summed to produce a final weighted score for each region. This approach reflects the reality that development decisions require multiple favourable characteristics to be balanced simultaneously. A region with the highest mean wind speed, for example, may not represent the strongest overall candidate if turbulence, directional behaviour, or seasonal performance are materially weaker.

The weighted framework therefore provided a structured, transparent, and repeatable method for ranking candidate regions and selecting the preferred study area for detailed GIS screening.

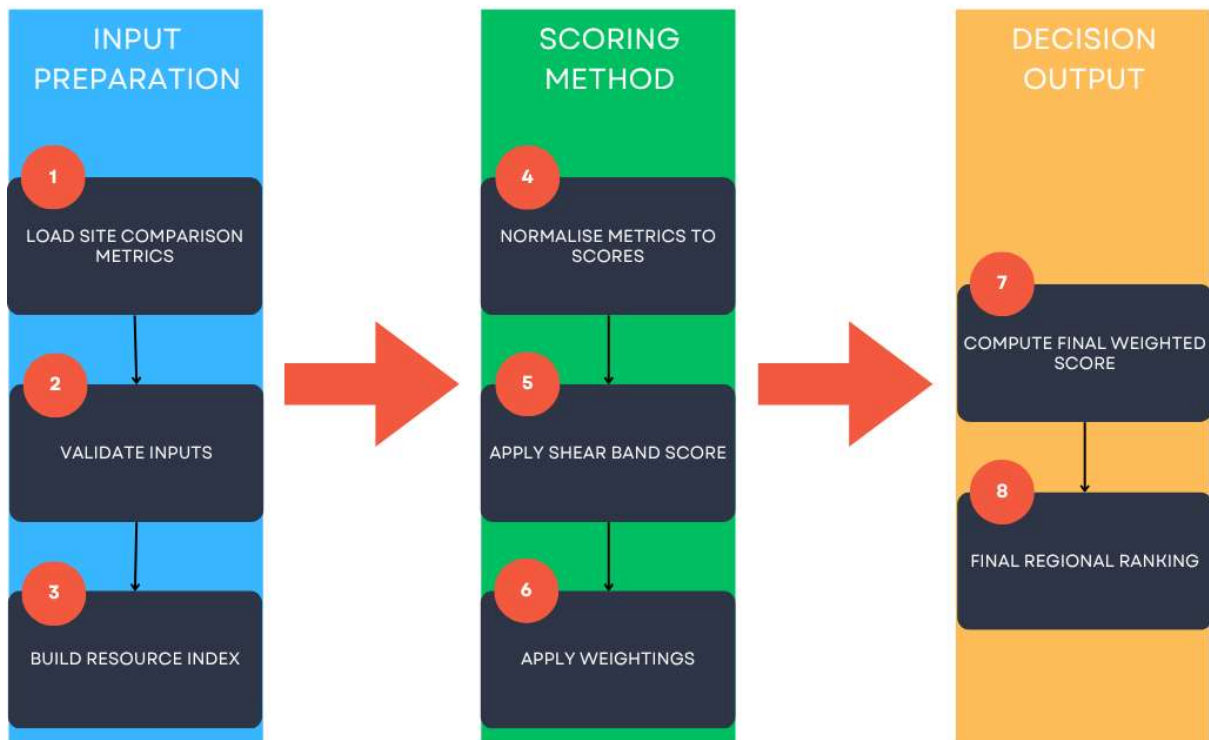


Figure 5 - Workflow for regional wind resource scoring, including metric normalisation, weighting, and final ranking.

ERA5-derived site metrics were validated, transformed into comparative component scores using metric-specific scoring rules, weighted according to engineering relevance, and aggregated into final candidate-region rankings.

3.6 Directional Wind Climate Considerations

Wind direction frequency is particularly important in wind farm engineering because wake interactions are highly directional. Turbine rows aligned poorly to the prevailing wind can materially reduce net energy yield.

For each region, dominant sectors were concentrated in the south-westerly to westerly range, consistent with Ireland's Atlantic exposure. This directional behaviour later informed layout interpretation and wake-loss analysis.

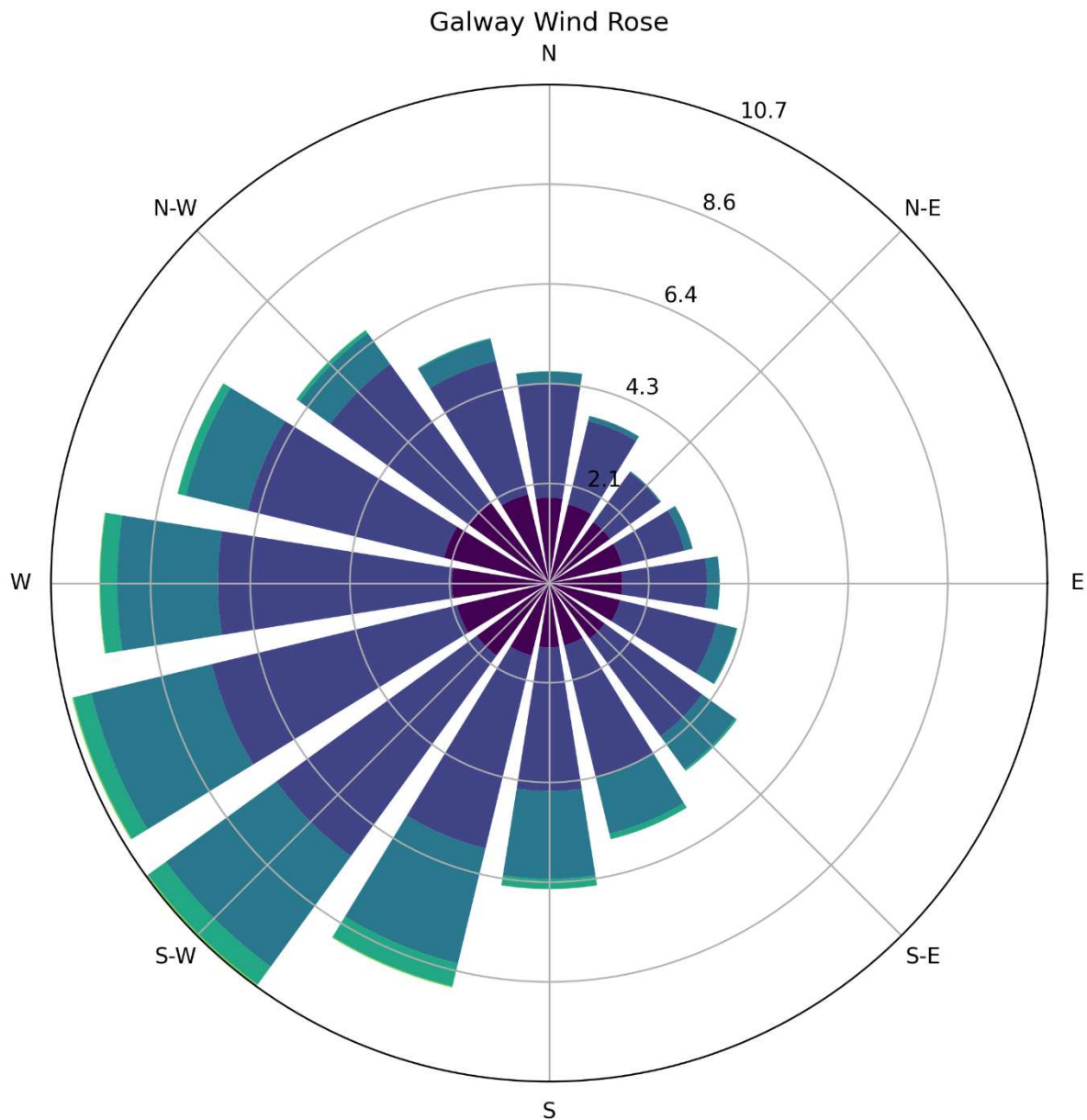


Figure 6 - Example wind rose for the Galway screening location, showing directional frequency distribution derived from ERA5 data. Dominant south-westerly to westerly sectors indicate a prevailing Atlantic wind regime.

3.7 Screening Outputs and Decision Gate

The output of the regional screening stage was a ranked shortlist of candidate regions supported by quantitative evidence. This created a formal decision gate before progressing into higher-effort geospatial and engineering analysis.

A structured decision gate is important because it prevents premature optimisation of suboptimal opportunities. By first identifying the strongest broad region, subsequent GIS and layout work could be focused where it had the highest potential value.

The numerical results of this screening process are presented in the following section.

Section 4. Wind Resource Screening Results

4.1 Overview

This section presents the outputs of the regional wind resource screening process described in Section 3. The purpose of the screening stage was to identify which candidate region offered the strongest overall combination of energy potential and development suitability before progressing to parcel-scale GIS assessment.

Five candidate regions were evaluated using a weighted multi-criteria framework derived from ERA5 metrics. The resulting ranking provided the first major decision point in the project workflow.

4.2 Detailed Metric Comparison

Mean Wind Speed and Power Density

Galway demonstrated the strongest performance across both mean wind speed and power density metrics. Since available wind power scales approximately with the cube of wind speed, relatively modest increases in mean speed can translate into materially higher energy potential.

Donegal and Mayo also performed strongly and formed a competitive second tier. Their scores indicate that multiple western Irish regions may support attractive wind development opportunities.

The Midlands, while lower in raw resource terms, remained useful as an inland comparator illustrating the trade-off between meteorological strength and potentially different land-use conditions.

Kerry recorded a strong wind climate but ranked lower overall once the full multi-metric weighting framework was applied.

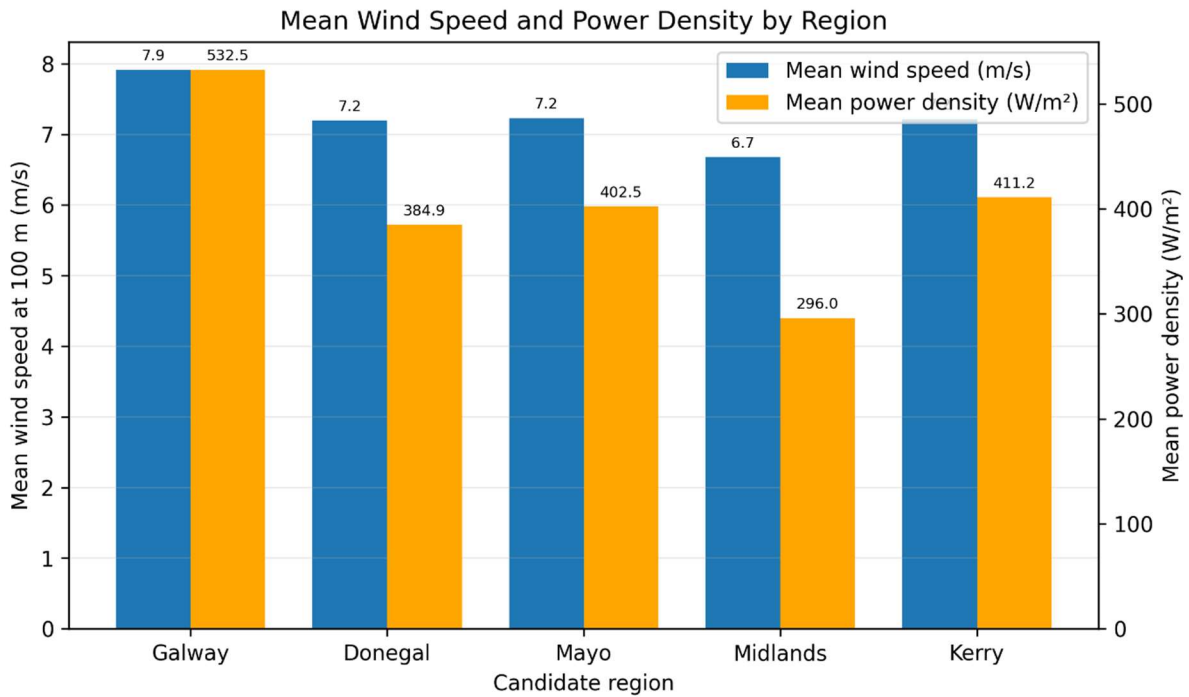


Figure 7 - Mean wind speed and power density comparison by region.

Weibull Shape and Scale Behaviour

The Midlands recorded the highest Weibull shape parameter (k), indicating a relatively stable wind speed distribution. This demonstrates why screening should not rely solely on average wind speed as some lower-wind sites may still offer desirable operational characteristics.

However, Galway's superior overall energy magnitude outweighed this advantage in the final ranking.

Table 5 - Weibull parameter comparison across candidate regions.

Region	Weibull Scale c (m/s)	Weibull Shape k	Interpretation
Galway	8.935	2.206	Highest scale parameter, indicating strongest characteristic wind speeds with balanced variability.
Donegal	8.12	2.288	Strong wind regime with relatively stable distribution. Good resource consistency.
Mayo	8.166	2.235	Competitive scale parameter with balanced shape factor. Strong overall wind climate.
Midlands	7.529	2.427	Lowest characteristic wind speed but highest shape factor, suggesting the most

			stable wind-speed distribution.
Kerry	8.144	2.156	Good characteristic wind speed but slightly lower shape factor, indicating somewhat broader variability.

Seasonal Performance

Galway also ranked highest in winter mean wind speed. This is strategically relevant because electricity demand is elevated during colder periods, increasing the potential system value of winter generation.

Strong winter output can therefore enhance the broader attractiveness of a project beyond annual average energy alone.

Directional Consistency

Directional concentration in Galway's south-westerly and west-south-westerly sectors suggested a coherent prevailing wind regime. This was an important positive indicator for later turbine layout work, where orientation relative to dominant inflow directions can influence wake losses.

4.3 Final Regional Ranking

The combined effect of all evaluated metrics was captured through the weighted scoring framework described in Section 3. The resulting regional ranking is presented in Table 6.

Table 6 - Final Regional Ranking.

Rank	Region	Weighted Score
1	Galway	3.836
2	Donegal	2.797
3	Mayo	2.774
4	Midlands	2.7
5	Kerry	2.291

Galway achieved the highest overall score and was therefore selected for detailed suitability screening and layout development.

4.4 Why Galway Ranked First

Galway's first-place ranking was driven by consistently strong performance across multiple categories rather than reliance on a single favourable metric.

Key indicators included:

- highest mean wind speed at 100 m (7.91 m/s);
- highest mean power density (532.5 W/m²);
- strongest winter mean wind speed (8.845 m/s);
- favourable Weibull scale parameter;
- strong directional concentration in prevailing sectors;
- acceptable turbulence and shear characteristics.

This profile is significant because robust projects typically emerge from balanced performance rather than isolated excellence. Galway combined strong raw resource magnitude with a wind regime suitable for later engineering optimisation. This combination of characteristics indicates both high energy yield potential and favourable conditions for subsequent layout optimisation

4.5 Interpretation of Competing Regions

Although Galway was selected, the remaining rankings also provide useful insight into the wider opportunity set.

Donegal

Donegal finished second with a strong overall wind climate and competitive weighted score. Its result confirms the high-quality wind resource available in north-western coastal Ireland.

Mayo

Mayo ranked a very close third and performed strongly across several metrics. The narrow score gap between Donegal and Mayo suggests that either region could warrant further study under a broader development portfolio approach.

Midlands

The Midlands ranked fourth. While weaker in pure wind resource terms, it demonstrated comparatively strong stability metrics. This highlights that inland regions should not be dismissed automatically, particularly where grid access, planning conditions, or land assembly may differ.

Kerry

Kerry ranked fifth within this specific scoring framework. This does not indicate poor wind conditions, but rather that its combined metric profile was less favourable within the defined scoring framework.

4.6 Decision to Progress Galway to Detailed Screening

As seen in the rankings, Galway represented the strongest candidate for continued development analysis. Advancing Galway into parcel-scale GIS screening ensured that subsequent effort was focused on the highest-value opportunity identified during the national comparison stage.

This decision demonstrates an important development principle: detailed engineering effort should be gated by structured early-stage option screening wherever practical.

4.7 Link to Next Stage

Having selected Galway as the preferred region, the next project stage assessed whether strong wind resource conditions could be translated into realistic developable land once practical constraints were applied.

The following section therefore moves from meteorological opportunity to physical site suitability through GIS-based land screening.

Section 5. GIS Constraints and Land Suitability Methodology

5.1 Overview

Selection of a strong wind resource region does not guarantee that a viable wind farm can be developed. Many high-wind areas contain physical, environmental, planning, or access constraints that materially reduce usable land area. For that reason, regional screening must be followed by a detailed land suitability assessment.

This section describes the GIS-based methodology used to convert the selected Galway study area from a broad meteorological opportunity into a mapped inventory of potentially developable land parcels.

The objective of this stage was to identify land that remained potentially suitable after applying practical exclusion constraints, and to create a structured parcel dataset for later engineering screening.

5.2 Role of GIS in Early-Stage Wind Development

Geographic Information Systems (GIS) are widely used in renewable energy development because they allow multiple spatial constraints to be evaluated simultaneously. Rather than relying on qualitative judgement alone, GIS workflows can apply consistent rules across large areas and produce transparent outputs that can be audited or refined later.

For wind development, GIS screening is particularly valuable during the desk-study phase because it helps answer three critical questions:

1. Where is development clearly unsuitable?
2. Where might development remain possible subject to further study?
3. How fragmented or contiguous is the remaining land bank?

These questions directly affect turbine count potential, infrastructure practicality, and project risk.

5.3 Constraint Categories Applied

The Galway study area was screened using multiple exclusion layers representing typical onshore wind development constraints in Ireland. Buffer distances and thresholds were selected to reflect current planning guidance, environmental protection requirements, and the strategic direction of national climate and renewable energy policy. These values should be interpreted as feasibility-stage screening assumptions rather than definitive consenting limits, with detailed site-specific assessment required at later project stages. In practice, final setback distances and constraint interpretation would be refined through planning guidance, environmental assessment, and detailed site investigation.

The principal constraint categories were as follows.

Table 5 - Constraint layers applied during GIS screening.

Constraint Type	Source Dataset	Screening Purpose	Indicative Buffer / Rule Applied	Treatment
Building setback	High Value Dataset - Buildings	Reduce likely conflict with residential receptors and screen unsuitable land near occupied structures	1000 m setback buffer	Hard exclusion
Road setback	Roads - National 250k Map of Ireland	Avoid direct encroachment on roads and preserve practical separation for siting	100 m setback buffer	Hard exclusion
Watercourse setback	High Value Dataset - Water	Avoid direct encroachment on mapped water features and provide basic environmental protection margin	50 m setback buffer	Hard exclusion
SAC ecological constraint	NPWS Special Areas of Conservation	Exclude environmentally sensitive designated conservation areas	SAC boundary + 50 m buffer	Hard exclusion
SPA ecological constraint	NPWS Special Protection Areas	Exclude environmentally sensitive designated bird protection areas	SPA boundary + 50 m buffer	Hard exclusion
Slope exclusion	Copernicus GLO-30 DEM	Remove steeper terrain likely to increase civil engineering and constructability risk	Slope > 15° excluded	Hard exclusion

Multiple desk-study exclusion layers were combined to identify potentially developable land within the Galway study region. Buffer distances and thresholds represent feasibility-stage screening assumptions rather than final planning or design limits.

5.4 GIS Workflow Process

The screening methodology followed a sequential overlay process in QGIS. This allowed the study area to be progressively reduced as constraints were applied.

The core workflow was:

1. Define the selected Galway study boundary.
2. Import relevant geospatial constraint layers.
3. Apply buffers and exclusion zones where required.
4. Merge constrained areas.
5. Subtract excluded land from the study boundary.
6. Convert remaining land into individual polygons.
7. Clean topology and remove artefacts where necessary.

8. Export final parcel dataset for engineering scoring.

This structured approach ensured that the final parcel inventory was traceable to explicit geospatial rules rather than subjective visual selection. The sequential workflow described above is illustrated in Figure 8.

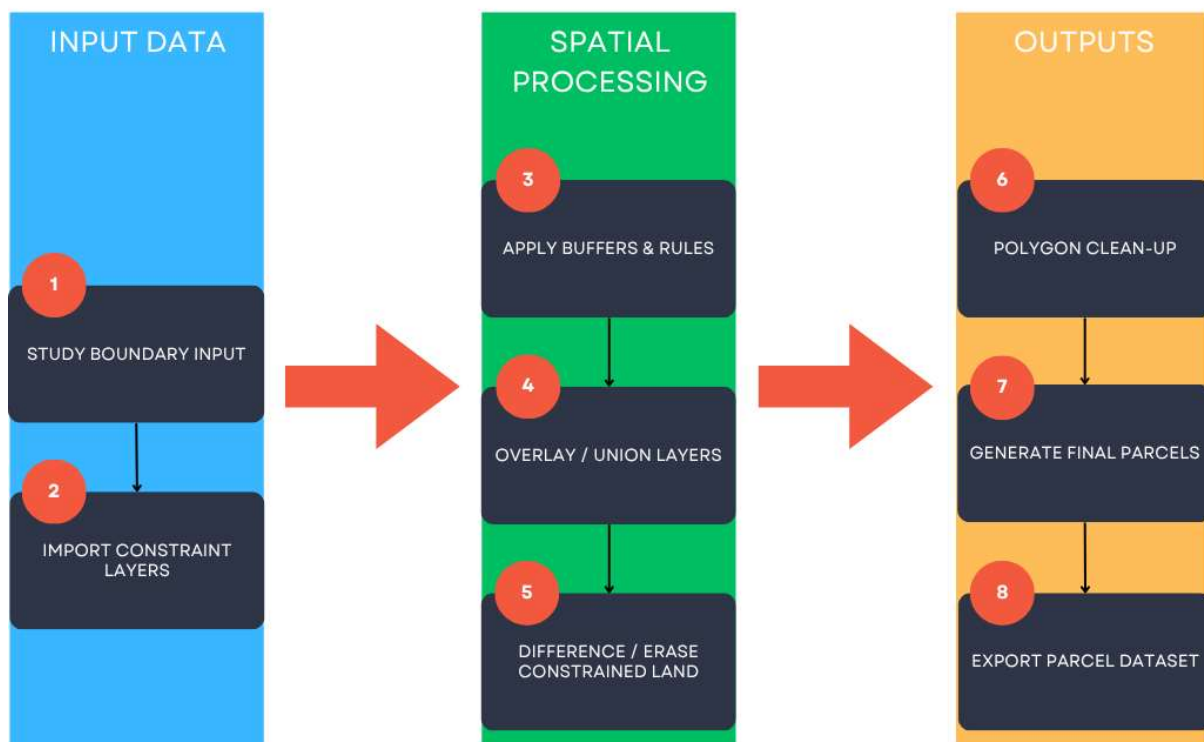


Figure 8 - GIS processing workflow used in QGIS.

5.5 Treatment of Fragmentation and Residual Land

A common outcome of multi-layer screening is fragmentation of the remaining land into many small residual polygons. While such fragments may be technically unconstrained in a narrow geometric sense, they are often too small, isolated, or irregular to support meaningful turbine development.

This distinction is important to make clear that suitability mapping does not automatically equal developability.

Accordingly, all remaining polygons were retained initially, but later engineering screening introduced parcel viability classes based on usable area and development practicality.

5.6 Creation of the Parcel Dataset

Following completion of the overlay process, the final unconstrained land bank was dissolved into discrete parcels. Each parcel was assigned a feature identifier (FID) and measured geometrically.

Attributes calculated for later screening included:

- parcel area in square metres;
- parcel area in hectares;
- parcel area in square kilometres;
- perimeter length;
- proximity to nearest building;
- proximity to nearest road;
- derived suitability indicators.

This created the working dataset used in the next stage of parcel prioritisation. To aid interpretation of the fragmented land pattern, the five largest remaining parcels are labelled in Figure 9, with FID 7 highlighted as the largest. Parcel size is an important initial indicator of potential development viability, although further screening is required before suitability can be confirmed.

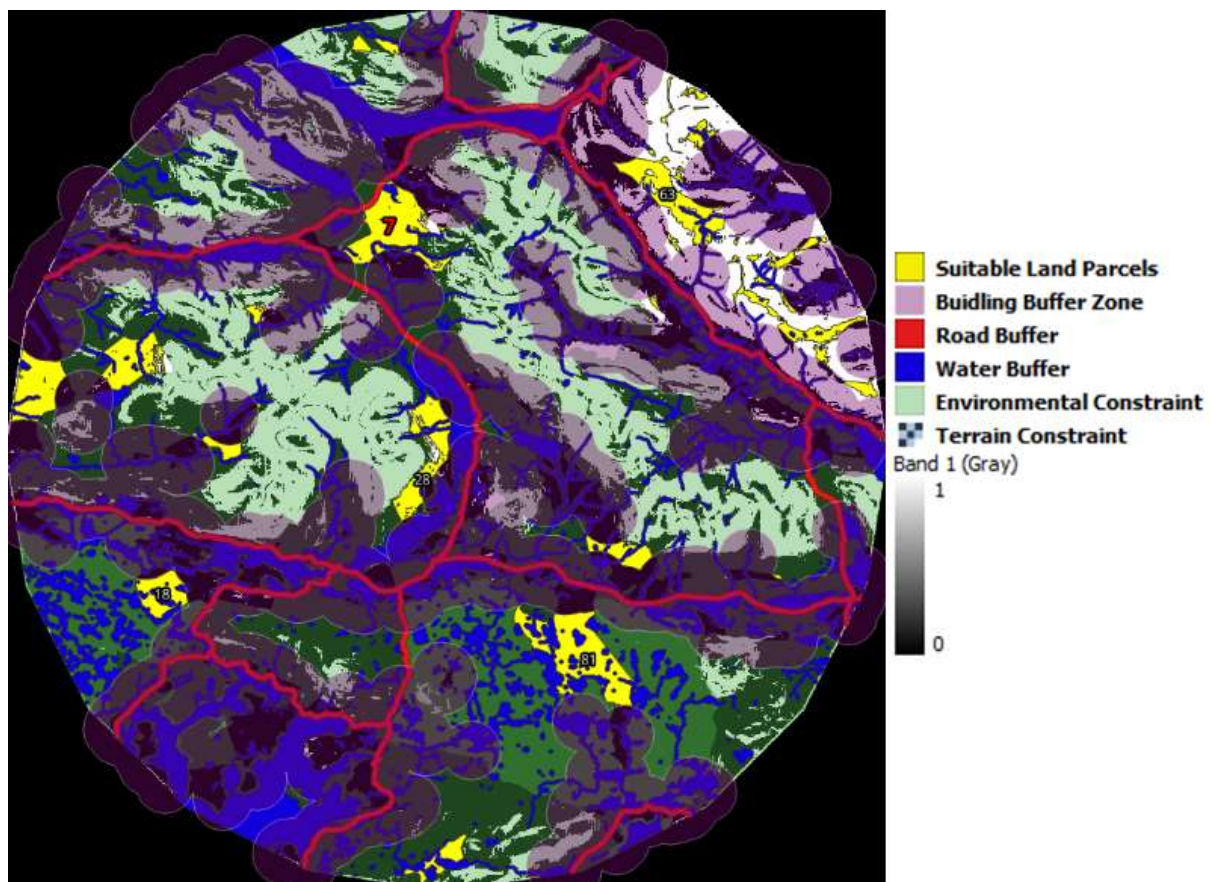


Figure 9 - Final parcel map after constraint filtering showing the five largest remaining parcels within the Galway study area. Parcel FID 7 is highlighted as the largest candidate.

5.7 Limitations of Desk-Based GIS Screening

Although highly valuable, GIS desk studies remain an early-stage approximation. They do not replace detailed site investigations or statutory assessments.

Important limitations include:

- buffer assumptions may differ from future planning requirements;
- public datasets may contain omissions or timing differences;
- terrain complexity may require higher-resolution modelling;
- land ownership and legal access are not captured automatically;
- ecological sensitivities may require field verification;
- construction constraints can emerge only during detailed design.

The outputs should therefore be interpreted as a screened opportunity set rather than confirmed developable land.

Once the Galway study area had been reduced to a parcel inventory, the next task was to determine which parcels were realistically suitable for development.

The following section presents the parcel scoring system used to classify viability, rank opportunities, and select the preferred development parcel for turbine layout optimisation.

5.8 Engineering Significance of GIS Screening

The GIS stage acts as the bridge between theoretical resource opportunity and realistic engineering development. A region may exhibit excellent wind conditions, but if suitable land is excessively fragmented, inaccessible, or constrained, practical project value can fall sharply.

Conversely, a slightly weaker wind region with cleaner land geometry can sometimes outperform a stronger but heavily constrained location once real design limitations are considered.

For this reason, the land suitability stage is not an administrative exercise. It is a core engineering filter that shapes all downstream layout and energy decisions.

However, while this stage identifies land that remains technically available after constraint filtering, it does not distinguish between parcels of differing engineering value. A further level of screening is therefore required to assess which parcels are realistically suitable for wind farm development.

Section 6. Parcel Screening, Viability Classification, and Site Selection

6.1 Introduction

Building on the GIS-derived parcel dataset, the next stage of the project translated spatial suitability into engineering-level site selection. While the GIS stage identified land that remained after exclusion constraints were applied, suitability at this point was purely spatial. A parcel may remain unconstrained on a map while still being too small, too fragmented, too isolated, or too impractical to support a utility-scale wind development.

Accordingly, a second-stage parcel assessment methodology was developed to move from binary spatial suitability toward engineering relevance. This stage introduced parcel viability classes based on available area, followed by a priority scoring system that considered proximity to roads and buildings. The intention was not to replace detailed engineering design, but to provide a transparent and repeatable framework for narrowing a large set of candidate polygons into a realistic shortlist for turbine layout optimisation.

The parcel selection process was particularly important because the screened Galway study area contained a large number of residual polygons of varying size and quality. Many represented small fragments created by overlapping buffer constraints or irregular edge remnants with little realistic development value. Others were large enough to justify detailed study. A structured ranking system therefore allowed effort to be concentrated on the parcels most likely to support an economically and technically credible wind farm.

6.2 Candidate Parcel Population After Constraint Filtering

After all desk-study constraints and project-area boundaries were applied, the final screened dataset contained 214 individual land parcels. These parcels varied significantly in geometry, area, and accessibility characteristics.

The resulting parcel population broadly fell into three categories:

1. Large contiguous parcels with sufficient area to potentially host multiple turbines and associated infrastructure.
2. Medium parcels that may support limited development or future expansion depending on geometry.
3. Small fragmented remnants created by overlapping exclusion zones and therefore of limited standalone value.

This distribution is typical of exclusion-based screening exercises. Although many polygons remain technically unconstrained, only a small subset generally merit progression into engineering layout studies.

6.3 Parcel Viability Classification by Area

The first screening layer introduced an area-based viability classification. Available land area is one of the strongest early indicators of whether a parcel can support commercial-scale wind development, particularly once turbine spacing, internal setbacks, access routes, crane pads, and micro-siting flexibility are considered.

Three parcel classes were defined:

Class A - Primary Candidate

Parcels larger than 2.5 km². These were considered capable of supporting substantial turbine development and therefore prioritised for direct optimisation.

Class B - Secondary Candidate

Parcels larger than 1.0 km² and up to 2.5 km². These parcels may support smaller schemes, phased expansion, or supplementary turbines depending on geometry.

Class C - Fragment / Low Priority

Parcels of 1.0 km² or less. These were generally considered too constrained or too small for primary standalone development, though some may retain local strategic value.

The classification thresholds were selected to reflect practical development scale rather than purely mathematical breakpoints. Their purpose was to rapidly separate realistic utility-scale candidates from parcels unlikely to justify further modelling effort. The spatial distribution of these viability classes across the study area is shown in Figure 10.

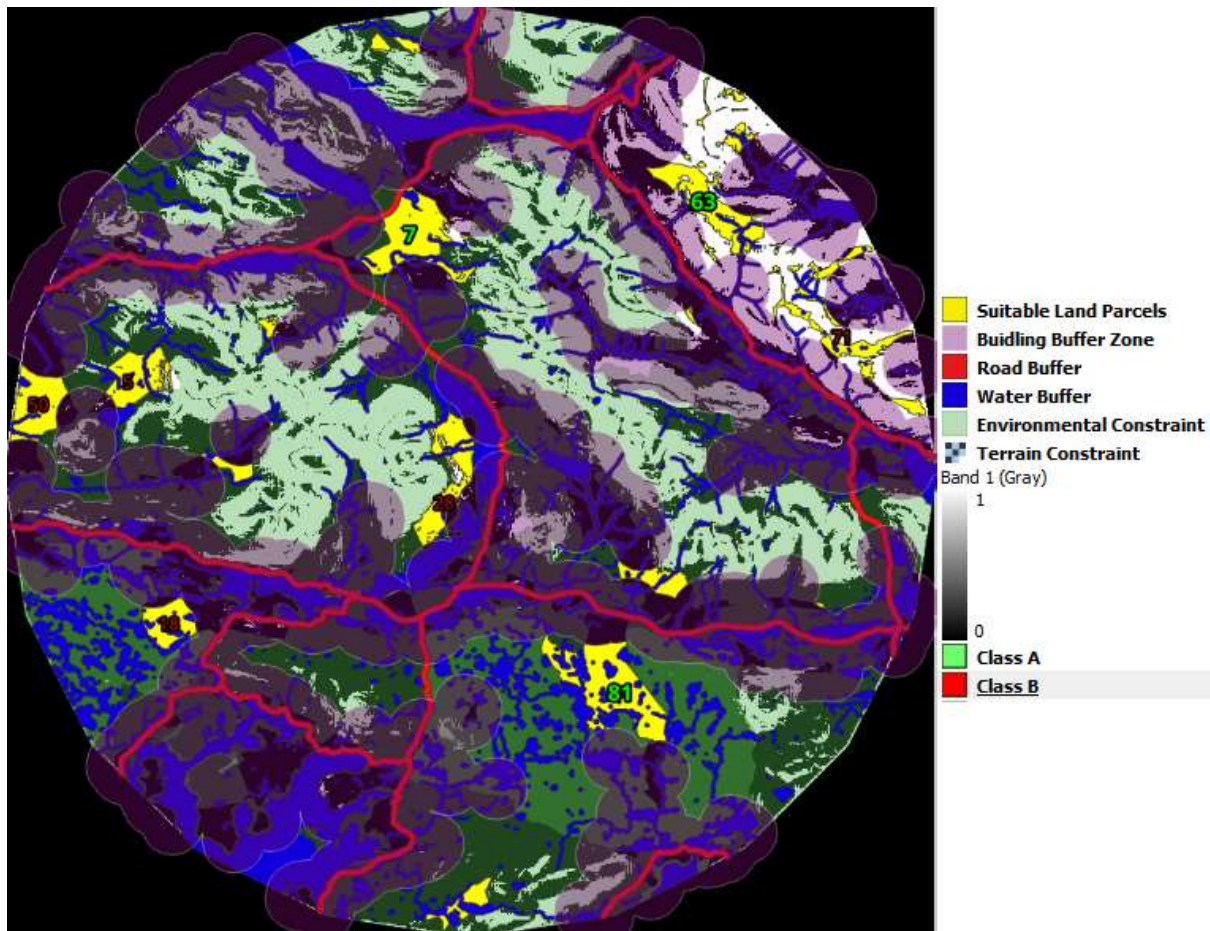


Figure 10 - Parcel viability classification across the Galway study area following GIS constraint filtering. Class A (green) parcels represent primary development candidates ($>2.5 \text{ km}^2$), Class B (red) parcels represent secondary candidates ($1.0\text{--}2.5 \text{ km}^2$), and smaller fragmented parcels are excluded from further consideration.

6.4 Priority Scoring Methodology

Area alone does not fully determine site attractiveness. A large parcel may still be difficult to develop if remote from road access or located close to residential receptors. Conversely, a moderately sized parcel with favourable access characteristics may offer stronger real-world development potential.

To account for this, a parcel priority score was introduced using two readily interpretable spatial indicators:

- Distance from buildings (positive when farther away). Greater separation from buildings reduces likely residential constraint risk, noise sensitivity, visual impact pressure, and planning conflict.
- Distance from roads (positive when closer). Reasonable proximity to road infrastructure may improve constructability, reduce new access requirements, and lower balance-of-plant complexity.

For each metric, parcel values were normalised across the screened parcel population to allow relative comparison. Building distance was scored on an increasing basis (farther = better), while road distance was scored on a decreasing basis (closer = better). The combined result produced a relative parcel priority score used to distinguish otherwise viable candidates.

The score was not intended as an absolute measure of project viability or planning consent likelihood. Rather, it functioned as a transparent desk-study ranking tool within the screened candidate pool.

6.5 Ranked Candidate Results

Application of the viability classes and priority scoring rapidly differentiated the candidate pool. The highest-ranked parcels are summarised in Table 8.

Table 6 - Highest-ranked candidate parcels from GIS suitability screening.

FID	Area (km ²)	Viability Class	Building Distance (m)	Road Distance (m)	Priority Score
7	3.545	A	970.4	99.3	0.921
18	1.527	B	975.2	631.9	0.659
63	2.683	A	978	702	0.627
81	3.906	A	979.7	776.2	0.592
28	1.674	B	965.2	856.2	0.537

FID 7 achieved a priority score of approximately 0.92, significantly higher than the next-ranked candidate at approximately 0.63. This separation is significant because it indicates that FID 7 was not merely the marginal winner of several similar parcels, but a clearly superior candidate under the selected screening logic.

FID 7 combined several desirable characteristics:

- Large area (3.545 km²)
- Strong separation from nearby buildings (~970 m)
- Very good road proximity (~99 m)
- Contiguous geometry suitable for optimisation
- Central developable area with scope for multiple layout iterations

The next-ranked parcels included several other Class A and Class B candidates, but none matched the combined access and setback balance achieved by FID 7.

6.6 Selection of FID 7 as Primary Development Parcel

Based on the combined ranking outputs, FID 7 was selected as the principal parcel for detailed layout development and wake modelling.

This decision was not based solely on area or solely on access, but on the integrated balance of metrics and geometry. The parcel offered sufficient land area for meaningful turbine deployment while remaining comparatively well positioned relative to access infrastructure and sensitive receptors.

Subsequent engineering assessment confirmed that FID 7 could support a feasible constrained layout of ten turbines internally, with later neighbouring parcel integration used to test whether the original

twelve-turbine project scale could be recovered more efficiently than forcing excessive density within a single polygon.

This reinforces a key development principle: the optimal site is not necessarily the largest, but the one that best balances resource, geometry, access, and constraint exposure.

6.7 Engineering Interpretation of Parcel Selection Results

The parcel ranking framework illustrates how geospatial data can be converted into project decisions through transparent engineering logic. Rather than selecting a site subjectively, the workflow documented each step from national wind screening to regional filtering, parcel classification, scoring, and final candidate selection.

This approach provides several practical benefits:

- repeatability if assumptions change;
- clear audit trail for decision-making;
- efficient targeting of modelling effort;
- ability to justify selections to technical and non-technical stakeholders;
- extensibility for future planning, cost, or environmental criteria.

It also reflects a broader engineering capability demonstrated by the project: where specialist commercial software workflows were unavailable, a bespoke but defensible site-selection pipeline was created using open tools and structured reasoning.

6.8 Limitations

The parcel ranking stage remained a desk-study screening tool and should be interpreted accordingly. Additional factors not explicitly included at this stage may materially influence final development viability, including:

- land ownership and land agreements;
- geotechnical suitability;
- peat stability or ground conditions;
- ecological survey findings;
- grid connection distance and capacity;
- planning landscape sensitivity;
- abnormal load route constraints;
- detailed civil engineering costs.

These factors are typically introduced in later feasibility and development phases.

6.9 Section Conclusions

From a starting pool of 214 screened parcels, a structured viability and scoring framework identified FID 7 as the strongest development candidate within the Galway study area. The parcel materially outperformed competing candidates across the selected scoring metrics and provided sufficient scale for detailed optimisation.

This established the basis for the next stage of converting a preferred parcel into practical turbine layouts, testing capacity limits, and quantifying wake-performance trade-offs under constrained development conditions.

Section 7. Turbine Layout Development and Engineering Design Logic

7.1 Introduction

Following identification of FID 7 as the preferred development parcel, the project moved from land screening into physical wind farm design. This stage moved from a site suitability question to answering, how many turbines could be credibly deployed within the selected area while maintaining practical spacing, respecting parcel geometry, and preserving acceptable wake performance.

Wind farm layout design is a multidisciplinary optimisation problem involving aerodynamic, spatial, and engineering constraints. Turbines must be positioned to maximise energy capture while limiting aerodynamic wake interaction, avoiding constrained boundaries, retaining constructability, and maintaining sufficient flexibility for future micrositeing. In commercial practice, this stage is often supported by proprietary optimisation platforms. In this project, an equivalent transparent workflow was developed using open-source methods and iterative engineering judgement.

The resulting layout methodology combined spacing heuristics, parcel-fitted geometry, repeated candidate generation, and wake-model performance testing. This enabled the project to move beyond theoretical site selection and produce implementable turbine arrangements supported by quantified energy outputs.

7.2 Design Basis and Turbine Technology Assumption

A single reference turbine platform was adopted throughout the optimisation stage to ensure comparability between all candidate layouts.

Selected turbine model: Vestas V136-4.2 MW

Rated power: 4.2 MW

Rotor diameter: 136 m

Hub height: 112 m

The turbine power and thrust characteristics were represented using engineering proxy curves suitable for feasibility-stage wake modelling. While not a substitute for OEM-certified performance data, this assumption provided a consistent basis for comparative design assessment across all candidate layouts.

The selected turbine scale is representative of modern onshore utility deployment and sufficiently large to illustrate realistic spacing, wake, and capacity trade-offs.

7.3 Baseline Layout Reference

Before constrained optimisation, the project retained a benchmark layout. The 12-turbine $5D \times 7D$ baseline layout was used as an unconstrained benchmark rather than as a final design target. This spacing basis provided a defensible engineering reference case because it represents a conventional preliminary wind farm spacing assumption, with tighter lateral separation and greater downwind

separation to account for wake recovery. The 12-turbine count also established a consistent 50.4 MW reference project scale using the selected 4.2 MW turbine. This allowed later constrained layouts to be assessed against a clear benchmark for installed capacity, wake loss, and net annual energy production.

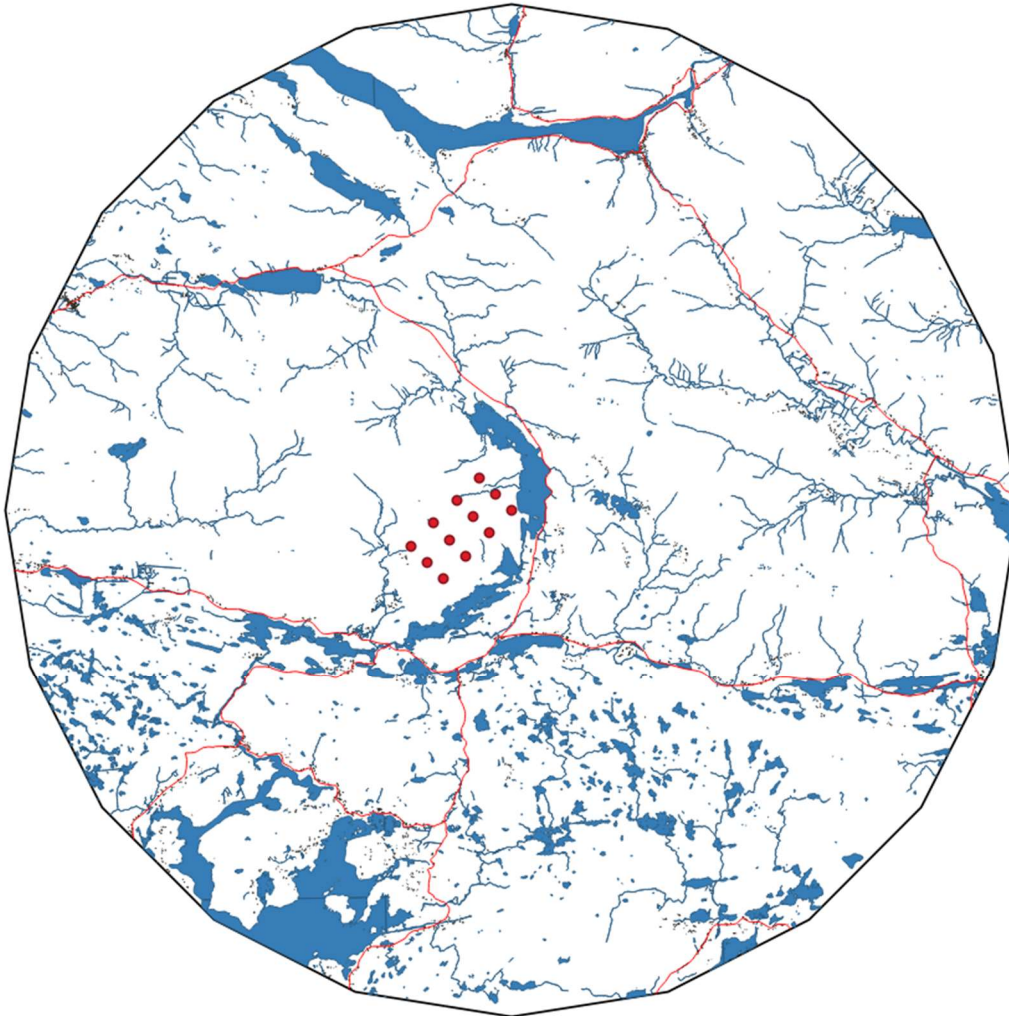


Figure 11 - Idealised reference layout positioned at the ERA5 sampling location, used to benchmark energy performance against parcel-constrained designs.

Its purpose was to answer two key questions:

1. What energy yield might be achieved under simplified unconstrained deployment assumptions?
2. How much performance is lost when the layout is redesigned around real parcel boundaries and practical development constraints?

Table 7 - Baseline reference layout.

Metric	Value
Turbines	12
Installed capacity	50.4 MW
Spacing basis	5D × 7D
Baseline minimum nearest spacing	680.2 m
Role in study	Benchmark comparator

7.4 Parcel-Constrained Layout Philosophy

Once the design process moved into FID 7, the unconstrained grid concept was no longer appropriate. The parcel geometry was irregular, with edge effects, internal constraints, and non-rectangular development zones. Forcing a rigid array into such a boundary would create inefficient turbine placement and elevated wake losses.

Instead, the project adopted a parcel-fitted layout philosophy based on the following principles:

- prioritise usable central land over constrained margins;
- align rows with prevailing wind logic where practical;
- maintain workable turbine separation distances;
- allow staggered rather than rigid row placement;
- preserve construction access flexibility;
- avoid over-densification solely to maximise turbine count.

This approach mirrors real engineering practice, where practical layouts are frequently adaptive responses to terrain and land control rather than mathematically perfect grids.

7.5 Iterative Capacity Testing Within FID 7

The performance of representative internal layout configurations was evaluated across a range of turbine counts to understand how increasing density influences energy yield and wake interaction. The results of this capacity sweep are summarised in Table 10, which presents gross and net annual energy production alongside wake loss for layouts ranging from five to twelve turbines within FID 7.

Table 8 - 5T-12T Internal Capacity Sweep

Layout	Turbines	Capacity (MW)	Gross AEP (GWh)	Net AEP (GWh)	Wake Loss (GWh)	Wake Loss (%)	Net Capacity Factor
5T	5	21	57.89	56.45	1.44	2.49	30.69%
6T	6	25.2	69.47	66.59	2.88	4.15	30.16%
7T	7	29.4	81.04	78.66	2.39	2.95	30.54%
8T	8	33.6	92.62	88.92	3.7	3.99	30.21%

9T	9	37.8	104.2	99.07	5.13	4.92	29.92%
10T	10	42	115.78	109.61	6.17	5.33	29.79%
11T	11	46.2	127.36	119	8.35	6.56	29.40%
12T	12	50.4	138.93	128.43	10.5	7.56	29.09%

The results show a clear trade-off between installed capacity and aerodynamic efficiency. While increasing turbine count raises total energy output, wake losses increase progressively as spacing becomes more constrained. Lower-density layouts (5T–7T) exhibit minimal wake interaction but underutilise the available land, whereas higher-density layouts (11T–12T) experience significantly greater wake losses due to reduced spacing. The ten-turbine configuration represents a balance between these effects, retaining strong energy output while avoiding the disproportionate efficiency penalties associated with further densification.

This capacity sweep therefore establishes a clear decision boundary: beyond approximately 10 turbines, additional capacity is gained at the expense of disproportionately higher wake losses, indicating that further internal densification is not an efficient design strategy.

7.6 Spacing Metrics and Engineering Interpretation

Turbine spacing was assessed using nearest-neighbour and pairwise separation distances expressed in both metres and rotor diameters (D). These metrics are widely used because wake recovery is strongly related to downstream spacing and crosswind separation.

For the preferred internal ten-turbine layout (v2):

- Minimum nearest spacing: 504.9 m
- Mean nearest spacing: 618.5 m
- Maximum nearest spacing: 885.2 m
- Minimum nearest spacing: 3.71D
- Mean nearest spacing: 4.55D

These values indicate a relatively compact but still workable onshore arrangement within a constrained parcel. Representative spacing metrics for key internal layout candidates are summarised in Table 11.

Table 9 - Selected spacing metrics for internal FID 7 candidate layouts.

Layout	Turbines	Min Spacing (m)	Mean Spacing (m)	Min Spacing (D)	Mean Spacing (D)
10T Candidate	10	504.9	618.5	3.71	4.55
11T Candidate	11	478.7	563.9	3.52	4.15
12T Candidate	12	451.7	525.8	3.32	3.87
9T Candidate	9	523.8	675.3	3.85	4.97

Table 11 highlights the effect of increasing turbine density on spacing quality and their impact on wake recovery. At approximately $3.71D$, as seen in the preferred 10-turbine layout, turbines are positioned at the lower end of typical onshore spacing but still allow partial wake recovery before downstream interaction occurs. In contrast, spacing closer to $3.32D$ in the 12-turbine internal layout indicates a more congested arrangement, where wake deficits persist into downstream machines, increasing energy losses and turbulence exposure. These spacing reductions directly explain the wake-loss trend observed in Table 10, linking geometric constraint to aerodynamic performance.

The spatial arrangement of the selected ten-turbine configuration within FID 7 is illustrated in Figure 12. The layout reflects the constrained geometry of the parcel while maintaining workable turbine spacing and avoiding over-densification near parcel boundaries.

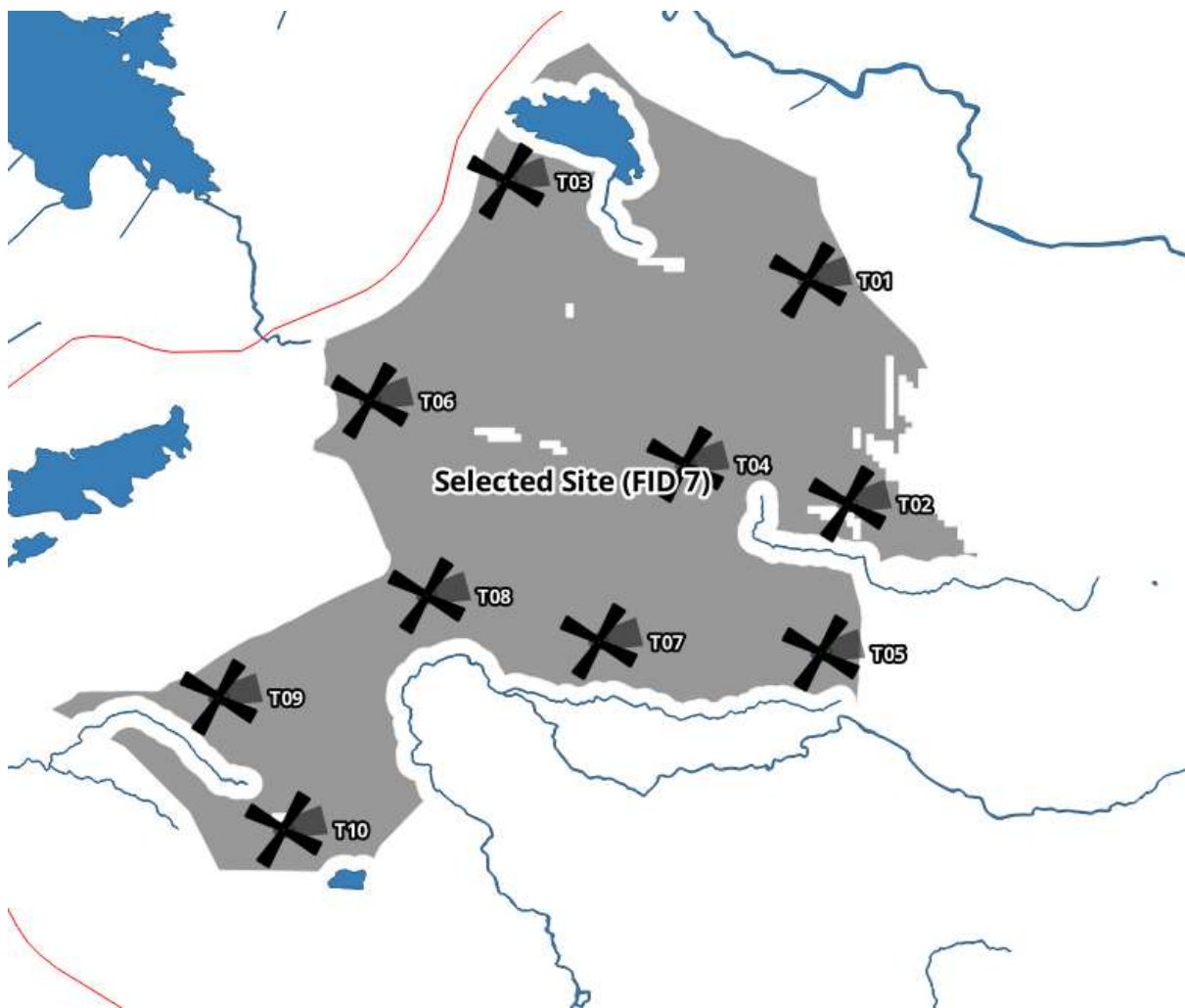


Figure 12 - Ten-turbine (10T) parcel-constrained layout within FID 7

7.7 Selection of the Preferred Internal Layout

Having identified the ten-turbine configuration as the most effective balance between capacity and wake loss from the initial capacity sweep, targeted refinement of turbine positioning was

undertaken. Several minor variations of the 10T layout were tested to assess whether small adjustments in spacing distribution could yield additional performance improvements.

Table 10 - Comparison of 10T layout refinement iterations

Layout Variant	Net AEP (GWh)	Wake Loss (GWh)	Wake Loss (%)	Net Capacity Factor	Change vs v1
10T v1	109.61	6.17	5.33%	0.2979	-
10T v2	109.74	6.04	5.22%	0.2983	0.11%
10T v3	109.65	6.12	5.29%	0.298	0.04%

As shown in Table 12, only marginal differences in performance were observed across the three layout variants. The maximum improvement in net AEP between configurations is approximately 0.11%, with similarly small variations in wake loss and capacity factor. In some cases, positional adjustments result in negligible or slightly negative changes in performance despite altered turbine spacing.

This indicates that the layout has effectively reached a near-optimal configuration under the assumptions and geometric constraints applied at this stage. The limited sensitivity of performance to further refinement suggests that additional desk-based optimisation is unlikely to yield meaningful gains. Instead, further improvements would require higher-fidelity modelling using site-specific wind data and more advanced wake interaction modelling.

On this basis, the 10T v2 configuration was selected as the preferred internal layout due to its marginally improved energy yield and reduced wake loss. This refined layout was subsequently used as the basis for multisite expansion testing, including the development of the selected Candidate D configuration.

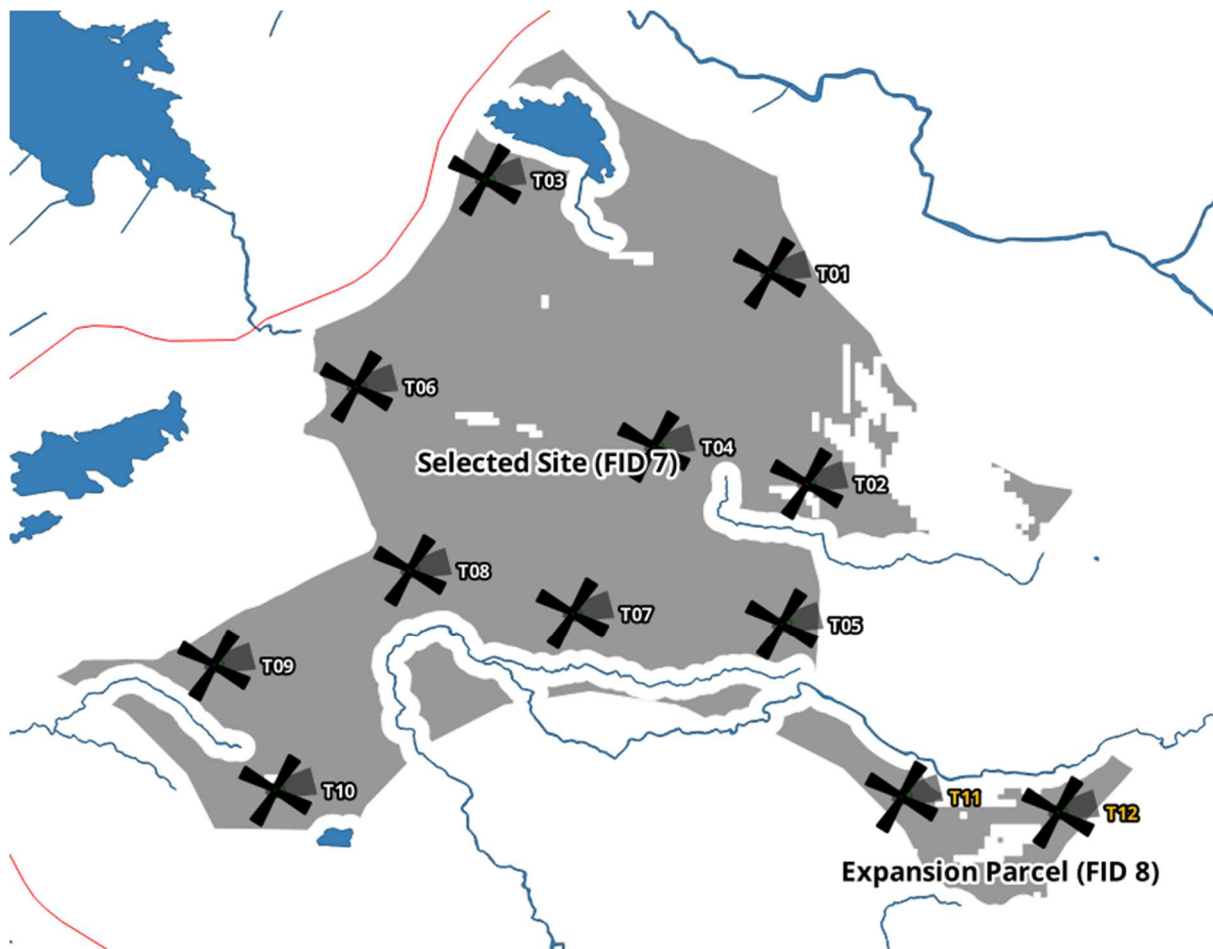


Figure 13 - Candidate D multisite layout combining the 10T internal configuration within FID 7 with two additional turbines in neighbouring parcel FID 8.

The figure shows the preferred configuration combining a 10-turbine internal layout within FID 7 with two additional turbines located in a neighbouring parcel (FID 8). The spatial separation of the additional turbines into FID 8 improves wake recovery relative to the fully internal 12T layout, illustrating the advantage of distributed development.

7.8 Rationale for Adjacent Parcel Expansion Over Internal Densification

Although a twelve-turbine benchmark existed, forcing all twelve turbines inside FID 7 proved less efficient than deploying ten turbines internally and locating two additional machines in nearby suitable parcels.

The surrounding area contained four smaller parcels close to FID 7. After applying the same perimeter buffering logic, two were eliminated and the remaining two parcels provided four feasible additional turbine positions. Combining these positions into mutually compatible pairs produced five multisite expansion scenarios (A-E after one conflict pair was removed).

This was an important engineering decision. Rather than preserving a headline turbine count at any cost, the design process prioritised whole-project performance. Spatial distribution across

neighbouring parcels improved separation opportunities and reduced the penalty associated with over-dense single-parcel development.

7.9 Engineering Capability Demonstrated by the Workflow

This section also highlights a broader project competency beyond conventional wind engineering calculations. In the absence of specialist commercial optimisation tools, the study developed a bespoke layout design pipeline that remained traceable and technically defensible.

The workflow demonstrated:

- translation of GIS outputs into design inputs;
- use of quantitative spacing rules;
- iterative scenario generation;
- comparative performance testing;
- engineering judgement under software constraints;
- communication of results in decision-ready form.

These are transferable engineering skills directly relevant to feasibility, development, and analytical roles where imperfect data and limited tooling are common realities.

7.10 Limitations

The layouts developed in this section should be interpreted as feasibility-stage concepts rather than final turbine coordinates. Detailed development would still require:

- topographic survey refinement;
- geotechnical and peat assessment;
- noise and shadow flicker compliance checks;
- transport route design;
- electrical balance-of-plant design;
- landowner negotiation and turbine micro-siting;
- bankable energy yield assessment using validated datasets.

Accordingly, the value of this stage lies in option screening and comparative engineering logic rather than final investment-grade design.

7.11 Section Conclusions

The project successfully converted a preferred screened parcel into a structured set of candidate wind farm layouts. Through iterative spacing-led design and comparative testing, a ten-turbine

internal configuration emerged as the best single-parcel solution, while adjacent parcel integration created a pathway to recover twelve-turbine project scale more efficiently.

This established the design basis for the next section: quantitative wake modelling, annual energy production analysis, and final configuration selection.

Section 8. Wake Modelling, Energy Yield Assessment, and Final Layout Selection

8.1 Introduction

Following development of candidate layouts, the next stage of the project focused on quantitative performance assessment. Spatially valid turbine layouts are not necessarily economically or energetically optimal. Their value must be tested through aerodynamic interaction modelling and annual energy production estimation.

For wind farms, wake effects are one of the most important internal design losses. As upstream turbines extract momentum from the flow, downstream turbines may experience reduced wind speed and increased turbulence intensity. This can lower annual energy yield, increase mechanical loading, and reduce the effectiveness of simply adding more turbines to a constrained site.

Accordingly, the project used wake modelling to compare candidate layouts on a consistent basis and identify the highest-performing developable configuration. This transformed the design process from geometry-led layout generation into evidence-led engineering selection.

8.2 Wake Modelling Framework

Wake simulations were undertaken using an open-source modelling workflow based on PyWake. This was selected as a technically credible alternative to commercial software packages and enabled full transparency of assumptions, repeatability of results, and direct integration with custom preprocessing scripts.

The modelling framework combined:

- turbine metadata and proxy power/thrust curves;
- site-specific wind climate inputs derived from ERA5 analysis;
- directional sector probabilities;
- Weibull wind speed parameters;
- candidate turbine coordinate layouts;
- automated batch comparison of multiple design cases.

Within this framework, PyWake computes annual energy production (AEP) by applying turbine power curves across the wind speed and directional distribution for each layout, with wake interactions modelled using the NOJ approach to derive both gross and net energy yields.

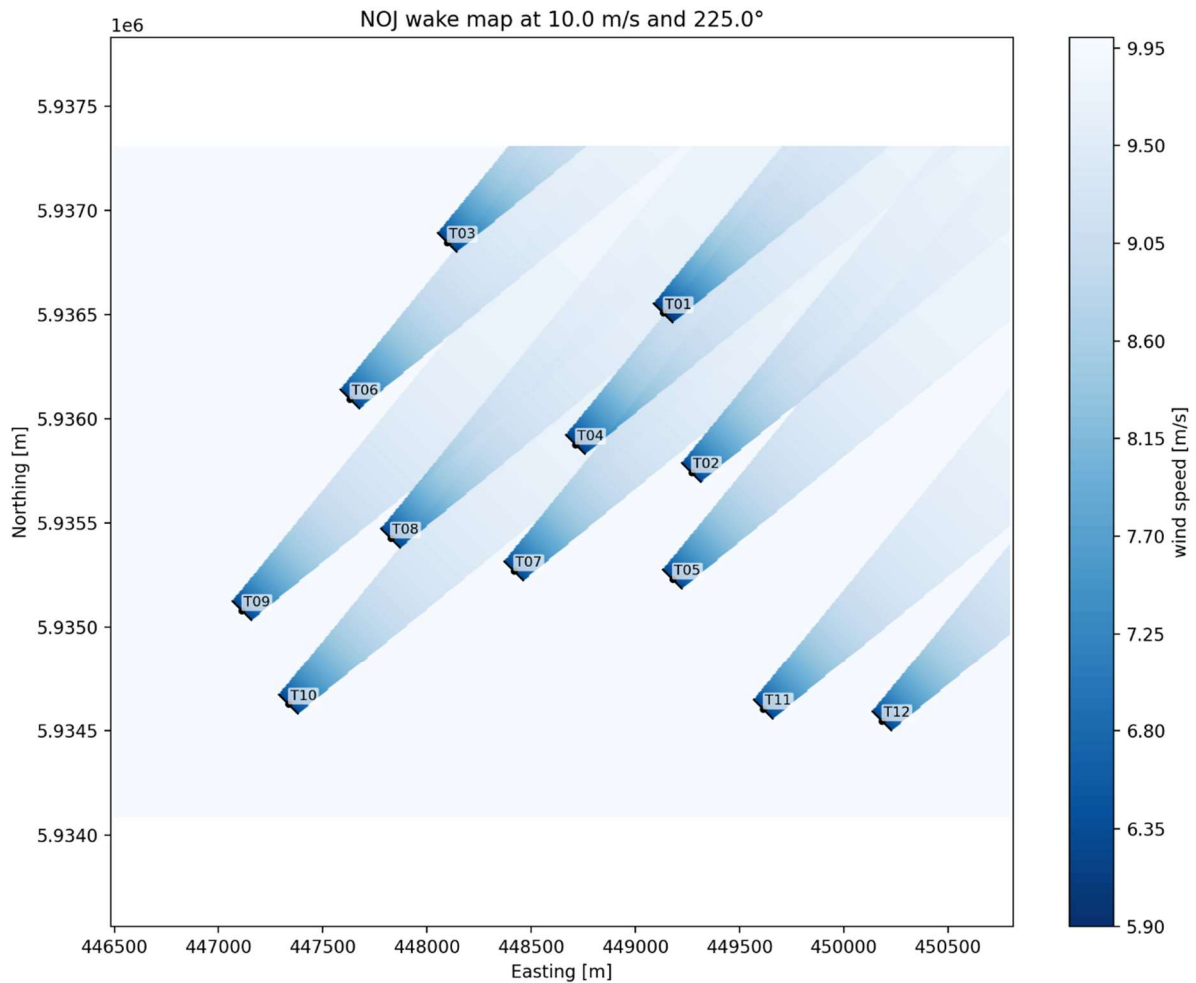


Figure 14 - NOJ wake map for the selected Candidate D layout at 10 m/s and 225° wind direction.

The figure illustrates the modelled downstream wake deficits used to estimate layout-specific wake losses and net AEP.

8.3 Wind Climate Inputs Used in Simulation

The selected Galway wind climate was used consistently across all candidate layout cases to ensure fair comparison. Directional probabilities and Weibull parameters derived from earlier ERA5 analysis formed the resource basis of the wake model.

Key simulation inputs included:

- Weibull shape factor (k): 2.206;
- Weibull scale factor (c): 8.935 m/s;
- Dominant direction 1: 225° (SW), probability 10.75%;
- Dominant direction 2: 247.5° (WSW), probability 10.58%;
- Dominant direction 3: 270° (W), probability 9.70%;

- Number of directional sectors: 16.

These values are consistent with a strong Atlantic south-westerly wind regime and reinforce the importance of directional spacing logic in layout design.

Table 11 - Wake model wind climate inputs for Galway.

Parameter	Value
Site	Galway
Directional sectors	16
Weibull k	2.206
Weibull c (m/s)	8.935
Dominant direction	SW (225°)
Dominant probability	10.75%
Secondary direction	WSW (247.5°)
Secondary probability	10.58%

Directional and statistical wind resource parameters used across all wake simulation cases. Inputs derived from ERA5-based site characterisation and applied uniformly for comparative layout assessment.

8.4 Performance Metrics Evaluated

Each candidate layout was assessed using a common performance set. This ensured that layout ranking was based on overall project value rather than a single headline metric.

The principal outputs were:

- Gross AEP (GWh): annual energy before wake losses;
- Net AEP (GWh): annual energy after wake losses;
- Wake Loss (GWh): energy reduction caused by internal wake effects;
- Wake Loss (%): percentage energy penalty relative to gross output;
- Capacity Factor: energy output relative to installed capacity;
- Energy per MW / per Turbine: useful normalised efficiency indicators.

Together, these metrics allow distinction between large layouts that merely install capacity and layouts that convert capacity into useful energy efficiently.

8.5 Internal FID 7 Candidate Results

Representative performance results for internal FID 7 layouts are summarised in Table 14.

Table 12 - Representative internal Fid_7 layout performance results.

Layout	Turbines	Capacity (MW)	Net AEP (GWh)	Wake Loss (%)
5T Candidate	5	21	56.45	2.49
7T Candidate	7	29.4	78.66	2.95
9T Candidate	9	37.8	99.07	4.92
10T Candidate	10	42	109.61	5.33
10T Candidate v2	10	42	109.74	5.22
12T Candidate	12	50.4	128.43	7.56

The internal FID 7 capacity sweep detailed in Section 7 established a clear performance pattern, with increasing turbine density leading to diminishing returns due to rising wake losses.

Building on this established performance pattern, targeted refinement of the ten-turbine configuration confirmed the 10T v2 layout as the preferred internal design, which was subsequently carried forward for multisite evaluation.

8.6 Multisite Expansion Scenario Testing

To preserve the original twelve-turbine project scale without excessive crowding inside FID 7, additional two-turbine combinations were tested using neighbouring parcels. Five valid multisite expansion options (A-E) were generated and evaluated by combining the preferred internal ten-turbine layout with two external turbine positions.

This changed the design problem from maximum density within one parcel to optimal distribution across multiple parcels.

All five multisite options outperformed the internal twelve-turbine crowded layout in wake-loss terms, demonstrating the value of distributed placement.

8.7 Final Candidate Ranking

The best overall performing layout was Candidate D, a multisite layout with 12 turbines on sites FID 7 and FID 8, which became the selected final configuration.

Its key outputs were:

- Turbines: 12;
- Installed capacity: 50.4 MW;
- Gross AEP: 138.93 GWh;
- Net AEP: 131.61 GWh;
- Wake loss: 7.32 GWh;
- Wake loss percentage: 5.27%;
- Net capacity factor: 29.81%.

Compared with the internal twelve-turbine layout, Candidate D retained the same installed capacity but materially reduced wake losses and increased net energy yield.

Table 13 - Ranked multisite twelve-turbine candidate results.

Rank	Layout	Net AEP (GWh)	Wake Loss (%)	Selection Status
1	Candidate D	131.61	5.27	Selected
2	Candidate E	131.5	5.35	Reserve
3	Candidate C	130.85	5.82	Alternative
4	Candidate B	130.7	5.92	Alternative
5	Candidate A	130.59	6	Alternative

Performance ranking of five multi-parcel expansion scenarios derived from the preferred internal ten-turbine base layout plus two additional turbine positions. Candidate D achieved the highest net AEP.

8.8 Comparison Against Baseline Benchmark

The original unconstrained twelve-turbine benchmark remained useful as a reference point. It represents an idealised reference case, not a claim that $5D \times 7D$ is the optimal spacing for the final constrained site. Relative to that case, the selected Candidate D layout showed only a modest reduction in net AEP despite needing to respond to real parcel constraints.

Comparison outputs indicate:

- Baseline benchmark net AEP: 132.26 GWh;
- Selected Candidate D net AEP: 131.61 GWh;
- Difference: -0.65 GWh (-0.49%);
- Wake loss increase: +0.46 percentage points.

This is an important result as it suggests that careful constrained engineering design recovered almost all of the benchmark energy potential despite moving from an idealised layout to a realistic developable arrangement.

Table 14 - Benchmark versus selected final layout.

Metric	Benchmark	Candidate D	Change
Turbines	12	12	0
Capacity (MW)	50.4	50.4	0
Net AEP (GWh)	132.26	131.61	-0.65
Wake Loss (%)	4.81	5.27	0.46
Net Capacity Factor (%)	29.96	29.81	-0.15

A comparison between the unconstrained benchmark, the single-parcel constrained layouts, and the selected multisite configuration is presented in Table 17. Results show limited performance sacrifice can be achieved after realistic redesign.

Table 15 - Summary of Key Layout Configurations

Layout Type	Turbines	Capacity	Net AEP	Wake Loss	Key Insight
10T v2	10	42 MW	109.74	5.22%	Best single-parcel
12T	12	50.4 MW	128.43	7.56%	Overcrowded
Candidate D	12	50.4 MW	131.61	5.27%	Selected multi-site
Benchmark	12	50.4 MW	132.26	4.81%	Idealised case

8.9 Interpretation of Wake Modelling Results

Several conclusions emerge from the modelling exercise.

First, turbine count alone is a poor optimisation objective. Dense layouts can increase gross capacity while eroding net value through wake interaction. This reinforces the importance of optimising for net energy yield rather than installed capacity alone, particularly in spatially constrained developments.

Second, parcel-constrained design does not necessarily imply major underperformance. With careful positioning, the final layout achieved near-benchmark energy output.

Third, adjacent parcel integration can be more valuable than forcing additional turbines into a single crowded site. This mirrors real-world development, where project boundaries and land agreements often evolve to improve performance.

Finally, open-source modelling can support credible engineering decisions when assumptions are transparent and results are interpreted appropriately.

From a development perspective, this demonstrates that careful layout design can preserve project value even under realistic land constraints, reducing the risk that early-stage site limitations materially impact final energy yield.

8.10 Limitations

The wake modelling performed here is appropriate for feasibility-stage comparison but should not be interpreted as a bankable energy assessment. Future refinement would typically include:

- higher-resolution wind resource datasets or on-site mast/LiDAR data;
- terrain-induced flow modelling;
- turbulence and fatigue load assessment;
- long-term loss factors and availability assumptions;
- electrical losses and curtailment modelling;
- uncertainty analysis (P50/P90 style assessments);

- OEM-validated turbine performance curves.

These refinements would normally accompany later development or financing stages.

8.11 Section Conclusions

The wake modelling stage identified the final project configuration through quantified comparison rather than assumption. The selected twelve-turbine multisite Candidate D layout delivered 131.61 GWh net annual energy, retained the full 50.4 MW project scale, and performed within 0.5% of the unconstrained benchmark.

This represents a strong engineering outcome: realistic site constraints were incorporated with only minimal sacrifice in performance. The next stage of the project can therefore move beyond technical layout feasibility toward grid integration, storage options, and techno-economic evaluation.

Section 9. Discussion, Key Findings, Limitations, and Future Development Pathway

9.1 Introduction

This section consolidates the outcomes of the project completed to date and interprets their broader engineering significance. The preceding sections moved sequentially from national wind resource screening, regional site selection, GIS constraint analysis, parcel prioritisation, turbine layout design, and wake-model performance assessment. Collectively, these stages demonstrate a complete early-phase wind farm feasibility workflow developed using transparent methods and publicly accessible tools.

The purpose of this final section is fourfold:

1. to summarise the principal technical findings of the study;
2. to interpret what those findings mean from a development perspective;
3. to acknowledge the limitations appropriate to a desk-study feasibility project;
4. to define the logical next steps for future project expansion, including grid, storage, and techno-economic assessment.

This structure is particularly important because the project is intended not only as a technical report, but also as a professional showcase of analytical capability, engineering judgement, and the ability to generate decision-ready outcomes under practical constraints.

9.2 Summary of the Project Workflow

The project established and executed a full screening-to-layout methodology for a conceptual onshore wind farm in Ireland. The workflow can be summarised as follows:

- national wind resource comparison using ERA5-derived metrics;
- ranking of candidate Irish regions using weighted engineering criteria;
- selection of Galway as the preferred regional study area;
- GIS exclusion mapping using desk-study constraints;
- conversion of screened land into ranked candidate parcels;
- selection of FID 7 as the preferred development parcel;
- iterative turbine layout generation across multiple capacities;
- wake modelling of internal and multisite configurations;
- final selection of a constrained 12-turbine layout with strong performance retention.

This sequence mirrors the decision structure used in real renewable development projects, where choices become progressively more detailed as uncertainty is reduced.

This structured progression from regional screening to layout optimisation reflects the staged decision-making process used in real-world wind farm development, where uncertainty is progressively reduced and engineering effort is focused on increasingly viable candidates.

9.3 Principal Technical Findings

9.3.1 Regional Wind Resource Selection

Among the candidate regions assessed, Galway ranked highest using the weighted resource methodology. It combined the strongest mean wind speed, highest power density, strong winter performance, and favourable directional characteristics.

This justified its progression into detailed spatial screening and later wake modelling.

9.3.2 Site Screening and Parcel Selection

After GIS constraints were applied, 214 residual parcels remained. Most were small fragments or low-value remnants, illustrating why binary suitability maps alone are insufficient for development decisions.

A second-stage viability and priority framework identified FID 7 as the strongest parcel. Its combination of area, road access proximity, and building separation made it the most credible development candidate.

This demonstrates that spatial screening alone is insufficient for development decisions, and must be complemented by engineering-based prioritisation to identify truly viable sites.

9.3.3 Layout Optimisation Findings

Internal capacity testing demonstrated that increasing turbine count inside a constrained parcel produces diminishing returns. Although higher-density layouts increased installed capacity, spacing quality deteriorated and wake losses rose.

The best single-parcel solution was a ten-turbine internal layout, while the best overall solution used neighbouring parcels to recover the original twelve-turbine project scale more efficiently.

This highlights a key development principle: maximising installed capacity does not necessarily maximise project value, particularly in spatially constrained environments.

9.3.4 Final Performance Outcome

The selected final layout (Candidate D) delivered:

- 12 turbines;
- 50.4 MW installed capacity;
- 131.61 GWh net annual energy production;

- 5.27% wake loss;
- a net capacity factor of approximately 29.8%.

This indicates that careful engineering design can preserve project viability even when transitioning from idealised layouts to realistic development constraints.

9.4 Current Study Limitations

In addition to data limitations, the methodology itself introduces simplifications that should be acknowledged. The use of ERA5 reanalysis data, simplified wake modelling assumptions, and geometric spacing heuristics means that results are best interpreted as comparative rather than absolute performance indicators.

Furthermore, the parcel scoring framework was intentionally simplified to maintain transparency and reproducibility. It does not capture all real-world development constraints, such as land ownership, grid connection feasibility, or planning risk. As such, the ranking should be interpreted as a screening tool rather than a definitive site selection outcome.

Future plans should also be noted, as with any desk-study feasibility exercise the results should be interpreted within scope.

The project does not yet include:

- grid connection availability or export capacity assessment;
- electrical design and collector system losses;
- storage system sizing or dispatch optimisation;
- capital expenditure and operating cost modelling;
- project finance metrics (NPV, IRR, LCOE);
- planning and consenting strategy analysis;
- ecological field surveys;
- noise, shadow flicker, or visual impact modelling;
- detailed ground investigation or peat stability review;
- bankable wind campaign measurement data.

These omissions are normal for an early-stage technical concept study and define the pathway for future work rather than undermine the current findings.

9.5 Future Development Pathway

To progress from feasibility-stage assessment to a development-ready project, a sequence of increasingly detailed technical studies would be required. These would refine both the energy yield estimate and the broader project viability under real-world constraints.

This project is intended to continue beyond the current report. The next logical modules would deepen the study from technical feasibility into project viability.

9.5.1 Grid Connection Feasibility

This stage would assess the feasibility of connecting the proposed development to the electricity network. It should examine:

- nearest substations and transmission assets;
- likely connection voltage level;
- route corridor constraints;
- curtailment exposure;
- reinforcement risk;
- indicative connection cost drivers.

9.5.2 Energy Storage Integration

Battery or hybrid storage options could then be assessed to determine whether storage improves project value through:

- output smoothing;
- constraint management;
- price arbitrage;
- ancillary services participation;
- improved export flexibility.

9.5.3 Techno-Economic Analysis

Once energy and grid assumptions are expanded, an economic model could estimate:

- CAPEX and OPEX ranges;
- annual revenue scenarios;
- sensitivity to power price assumptions;
- storage business case impacts;
- LCOE;
- NPV / IRR style metrics;
- risk-adjusted project attractiveness.

9.5.4 Final Development-Level Refinement

A later advanced phase could include higher-resolution resource modelling, refined turbine micro-siting, planning analysis, and investor-grade energy yield studies.

9.6 Final Conclusions

This project successfully demonstrated a complete early-stage wind farm development workflow from national resource screening to final layout selection. Using transparent and reproducible methods, a credible 50.4 MW conceptual wind farm was developed with an estimated net annual production of 131.61 GWh and performance close to an unconstrained benchmark.

Beyond the technical outputs, the work demonstrates broader professional capability in data-driven engineering judgement, geospatial analysis, custom workflow development, optimisation under practical constraints, and communication of complex findings to mixed audiences.

Overall, the project demonstrates that a structured, transparent, and reproducible engineering workflow can deliver credible feasibility-stage wind farm assessments using open-source tools, while still supporting meaningful development decisions.